PM2.5 Forecasting in Open-Pit Mining: Horizon-Dependent Predictive Value of Local Monitoring and Regional Atmospheric Information

Supplementary Methods

This supplement documents the mechanics behind the frozen parameters, criteria, and decisions reported in Methodology Sections 2.1–2.12.

S2.3 MERRA-2 temporal-alignment sensitivity scenarios (supports Section 2.3)

Three frozen sensitivity scenarios were retained alongside the primary area-weighted CST hour-end alignment: (i) nearest-cell spatial extraction with CST hour-end timing, which changed spatial support only; (ii) area-weighted extraction with CST hour-start labeling, a retrospective timing sensitivity in which the associated interval ended after the nominal label; and (iii) nearest-cell extraction with CST hour-start labeling, which changed both spatial and temporal assumptions. The hour-start scenarios matched 17,536 of 17,544 monitoring timestamps, leaving eight boundary records unmatched (versus nine under the primary hour-end convention). Scenario roles were frozen before fitting and were not selected using R^2, RMSE, MAE, test results, or SHAP values.

S2.4 Leakage-control verification (supports Section 2.4)

The leakage-control design combined five safeguards: target-time partition assignment; training-only estimation of caps and imputation values; uncapped held-out responses; backward-looking environmental features; and exclusion of future or target-time MERRA-2 measurements. Input identities, feature counts, timestamp continuity, split boundaries, and the absence of explicit future MERRA-2 fields were checked using machine-readable assertions before any model was fit.

Ordinary monitoring-variable missingness was measured before imputation. Missing counts were PM2.5 587, PM10 605, SO2 604, CO 594, NO2 592, O3 592, wind direction 55, wind speed 11, precipitation 10, temperature 3, relative humidity 3, and pressure 0. For every variable with a raw missingness indicator in the feature contract, that indicator was calculated from the unfilled source field and retained after the predictor itself was imputed. Thus imputation did not erase information about whether an observation had originally been unavailable.

The cap-then-median sequence first created the fixed training-calendar mask. The 99.5th-percentile cap for each designated pollutant was then estimated from nonmissing training observations and applied to a predictor-only copy across all partitions. Each capped predictor alias or uncapped direct predictor was subsequently selected as the imputation input, and its median was estimated on training rows only and used unchanged to fill missing values in training, validation, and test predictor inputs. Validation or test observations never contributed to a cap or median.

Predictor	Training-derived 99.5th-percentile cap	Validation values above cap	Test values above cap
PM2.5	223.070	0	2
PM10	352.108	26	0
SO2	121.260	3	0
CO	2.620	1	4

The MERRA-2 archive itself contained complete numerical values for all 731 daily subsets. Temporal alignment left nine monitoring timestamps without a matched MERRA-2 interval at the record boundary; these direct MERRA-2 inputs were filled with training-only medians, while merra2_origin_available preserved their unavailable status. After feature construction, history-edge NaNs from lag and rolling operations were left for CatBoost's native missing-value handling rather than filled with future or backcast values. Derived ratios or indicators were likewise permitted to remain NaN when their denominator or physical validity condition was not met.

Pressure had no formally missing source values, but three observations were recorded as 0.0 kPa and were physically invalid. In the executed pipeline, the ordinary model_input_pressure series retained those zeros; direct pressure lags, its 24 h rolling mean, the pressure–wind interaction, and winter/heating interactions therefore inherited the unfiltered values. A separate pressure_clean series masked values outside 80–110 kPa only for the high-pressure stagnation flag and 6/24 h pressure-tendency indicators. This produced exactly 12 non-edge tendency-indicator NaNs, which CatBoost handled natively. The asymmetric treatment is documented to reproduce the archived analysis and is not presented as an ideal physical-data cleaning rule.

A post-hoc validation-only sensitivity converted pressure outside 80–110 kPa to missing before the existing training-only median calculation and before any feature construction. It rebuilt both the archived-pressure and corrected-pressure inputs only through 7 August 2021 18:00, then refitted both model specifications at h1–h24 using the archived horizon-specific CatBoost parameters, original training targets, and original validation targets. Ninety-six fits were completed. The latest raw, target, and prediction timestamp equaled the validation endpoint; locked-test predictions and metrics remained untouched, and their SHA-256 hashes were unchanged before and after execution. Full results are provided in Supplementary Table S12.

PM2.5 targets were never imputed. A row was excluded whenever its horizon-specific target was missing or lay beyond the observed record. The 587 missing PM2.5 hours therefore removed $587 \times 24 = 14{,}088$ in-record origin–horizon pairs. The record endpoint removed another $1 + 2 + \ldots + 24 = 300$ pairs. Consequently, $17{,}544 \times 24 - 14{,}088 - 300 = 406{,}668$ eligible pairs remained across h1–h24.

S2.5 Full predictor construction rules (supports Section 2.5)

PM2.5 and PM10 histories included current values; 1, 3, 6, 12, and 24 h lags; missingness indicators; and 6 and 24 h rolling summaries (means, maxima, and standard deviations for PM2.5; means and maxima for PM10). NO2, SO2, and CO included current values, 1, 6, and 24 h lags, missingness indicators, and 24 h means. O3, temperature, relative humidity, pressure, and wind speed used current values, 1, 6, and 24 h lags, missingness indicators where defined, and 24 h means. Additional combinations included the PM2.5/PM10 ratio, PM10 minus PM2.5, the sum of NO2 and SO2, temperature × relative humidity, and pressure × wind speed.

Origin-time calendar variables included cyclic hour, day-of-year, and month encodings; month and season; winter, heating-season, and cold-season indicators; and winter-progress terms (winter = December–February, heating season = November–March, cold season = October–March). Selected local measurements and interaction terms were multiplied by winter and heating-season indicators to allow seasonally varying relationships.

The 23 engineered local-condition indicators were mechanistically motivated transformations of local variables only: wind direction sine/cosine components; calm, low, moderate, and high wind regimes; 6 h wind tendency; low-ventilation and stagnation conditions; 6 and 24 h pressure tendencies; humidity interactions with PM2.5, PM10, and temperature; current and accumulated precipitation; a recent wet-suppression indicator; the PM10/PM2.5 ratio; and coarse-particle indicators (PM10 minus PM2.5).

The MERRA-2 block's 73 predictors comprised, for each of the eight selected aerosol variables, current, 1, 6, and 24 h lagged values and a 24 h rolling mean (40 fields), one shared availability indicator (41st direct field), and current/rolling interactions with winter and heating-season indicators (32 seasonal-interaction fields). No MERRA-2 interaction was classified as a calendar feature. Nine target-calendar features were known deterministically at

93 forecast issuance: sine/cosine encodings of target hour, target day of year, and target month;
94 numerical target day of year; and target winter/heating-season indicators.

Supplementary Table S1. MERRA-2 aerosol variables retained and their use in the forecasting models.

MERRA-2 code	Atmospheric information represented	Unit	Predictors constructed for modeling	Intended predictive context
DUCMASS	Total dust aerosol column mass density	kg m−2	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Regional total mineral-dust loading
DUCMASS25	Dust PM2.5 column mass density	kg m−2	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Regional fine-dust burden relevant to PM2.5
DUEXTTAU	Dust extinction aerosol optical thickness at 550 nm	Dimensionless	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Optical loading associated with total dust
DUEXTT25	Dust-PM2.5 extinction aerosol optical thickness at 550 nm	Dimensionless	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Optical loading associated with fine dust
TOTEXTTAU	Total aerosol extinction optical thickness at 550 nm	Dimensionless	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Overall regional aerosol loading
SO4CMASS	Sulfate aerosol column mass density	kg m−2	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Regional secondary inorganic aerosol burden
BCCMASS	Black-carbon aerosol column mass density	kg m−2	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Regional combustion-related aerosol burden
OCCMASS	Organic-carbon aerosol column mass density	kg m−2	Current value; 1, 6, and 24 h lags; 24 h rolling mean; winter and heating-season interactions of current and rolling-mean values	Regional carbonaceous aerosol burden

S2.6 Hyperparameter search space (supports Section 2.6)

Persistence used the observed origin-time PM2.5 directly and therefore had no fitted hyperparameters. Month–hour climatology used training-period mean PM2.5 by target month and hour, with training-month and global-mean fallbacks. The fixed-parameter local-only and full-model CatBoost validation benchmarks used 400 iterations, depth 6, learning rate 0.05, L2 leaf regularization 5, RMSE loss, random seed 42, and all available processing threads.

The validation-only algorithm sensitivities used method-specific pipelines rather than CatBoost parameters. Ridge regression used median imputation, standardization, and alpha=10.0. Random Forest used median imputation, 60 trees, maximum depth 10, minimum leaf size 4, max_features=0.7, all available processing cores, and seed 42. Histogram gradient boosting used median imputation, 120 iterations, learning rate 0.06, 31 maximum leaf nodes, L2 regularization 2.0, and seed 42. These models were screening comparisons; the accepted tuned CatBoost specification remained fixed.

For the primary CatBoost models, the Optuna search space comprised 250–1,000 iterations, tree depths of 4–8, learning rates from 0.015 to 0.18 (logarithmic scale), L2 leaf regularization from 1 to 25 (logarithmic scale), random strength from 0 to 5, bagging temperature from 0 to 5, and border counts from 64 to 254. The objective was validation RMSE against uncapped PM2.5, with early stopping at 50 rounds per trial; the final iteration count was set to the best iteration plus one, with a minimum of 50. Optuna 4.9.0 used 20 startup trials and then adaptive TPE with deterministic per-search sampler seeds. Each of the 48 horizon-specification studies completed 50 trials, totaling 2,400 validation-only trials. The procedure does not establish globally optimal settings.

S2.8 Seasonal rolling-origin fold boundaries (supports Section 2.8)

The spring fold used data through 31 January 2021 for preprocessing training, February 2021 for validation, and March–May 2021 as the held-out test. The summer fold used data through 30 April 2021 for training, May for validation, and June–August for testing. The autumn fold used data through 31 July 2021 for training, August for validation, and September–November for testing. For each fold, predictor caps and imputation medians were re-estimated using that fold's own training history.

Complete winter coverage required two chronologically ordered folds. The 2020–2021 winter fold used January–November 2020 for training, December 2020 for validation, and January–February 2021 for testing. The 2021–2022 winter fold used data through October 2021 for training, November 2021 for validation, and December 2021 for testing. The accepted December predictions were reused without refitting, and out-of-fold January–February and December observations were concatenated (1,410 + 732 = 2,142 test targets per horizon); composite-winter R^2, RMSE, MAE, and bias were calculated directly from the pooled observed–predicted pairs rather than by averaging fold-specific R^2 values.

S2.10 Block-length estimation for the moving-block bootstrap (supports Section 2.10)

After predictions from the frozen models were available but before interval resampling, candidate block lengths were estimated using the positive-sequence integrated autocorrelation time of the full-model h24 residual series, accumulating autocorrelations until the first non-positive value (maximum search 168 h). Estimated lengths were 23 h for the main test, 20 h for spring, 28 h for summer, 25 h for autumn, and 11 h for the original winter series; the maximum value, 28 h, was then fixed for all main-test and seasonal interval calculations.

S2.12 Full feature-reduction procedure (supports Section 2.12)

The prospectively specified reduction analysis used 2,390 internal-selection observations per horizon (3 January–12 April 2021 after the 48 h embargo). Exact SHAP and 30-repeat, 28 h block-permutation importance were calculated using models fitted on development data with a fixed CatBoost configuration: 400 iterations, depth 6, learning rate 0.05, L2 leaf regularization 5, RMSE loss, and seed 42. For block permutation, the first 2,380 observations were reordered as 85 complete non-overlapping 28 h blocks while preserving order within each block. The final 10 observations remained unpermuted but were retained in every prediction and R^2 calculation. SHAP and positive permutation importance were normalized separately by horizon and summarized across the three bands. Separate 90%-cumulative SHAP and permutation collections were combined by union; repeat-specific permutation selection was recomputed 30 times, and features selected in at least 80% of repeats were classified as stable. Grouped permutations applied the same block reordering to all features in an eight-family block.

The subsequent disclosed comparative study audited 13 candidate ranking methods. Mutual information and F-regression represented filter methods; lasso, elastic net, Extra Trees, Random Forest, and native CatBoost represented embedded methods; RFE, RFECV, forward sequential selection, and custom family

elimination represented wrapper methods; and exact SHAP plus block permutation supplied model-specific evidence. Lasso (alpha=0.001) and elastic net (alpha=0.001, l1_ratio=0.5) did not meet the frozen convergence checkpoint and were excluded from all-horizon scaling. The final matrix therefore contained 11 methods: mutual information, F-regression, Extra Trees (250 trees, minimum leaf size 2), Random Forest (250 trees, minimum leaf size 2, max_features=0.7), native CatBoost using the fixed configuration above, RFE, RFECV, forward sequential selection, custom family elimination, exact SHAP, and block permutation. Wrapper methods used correlation-pruned representatives and three-fold expanding chronological validation with a Ridge (alpha=1) estimator; forward selection used a frozen 25-feature target.

Within each method and horizon, negative importances were set to zero for normalization and each method's positive normalized importance summed to one. At each horizon, the recorded normalized scores available for each feature were averaged; features absent from correlation-pruned wrapper inputs had no wrapper entry and were not assigned zero. The 59 highest-ranked features were retained separately at every horizon. Thus the budget was common, but feature identities were horizon specific. Consensus tiers, cumulative-90% candidates, fixed 75/100/125/150 budgets, band-specific candidates, and subsequent correlation-clustering configurations were evaluated separately and did not directly generate the final 59 lists.

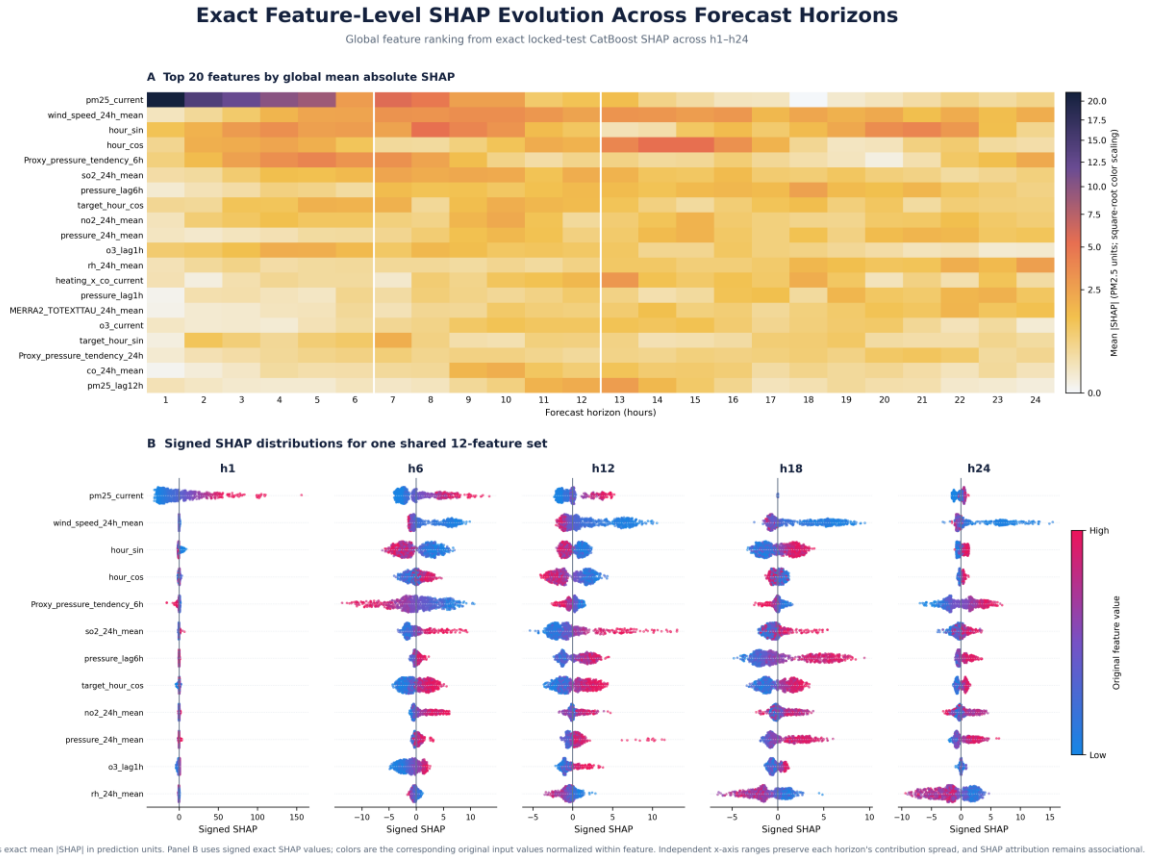
All reduced-versus-full validation refits used the same fixed CatBoost configuration rather than independently tuning each feature count. The post-hoc search evaluated a common 59-predictor budget and band-specific short-, medium-, and long-horizon alternatives under the disclosed joint 5% performance-loss limit. Stability testing used a paired non-circular 28 h moving-block bootstrap with 1,000 resamples drawn from all valid start positions and 30 deterministic refits (seeds 42–71) with only random_seed changed. The same resampled indices were applied to observed, full-model, and reduced-model series. The band-specific

boundary audit compared 14 versus 15 predictors at h6, 50 versus 51 at h10, and 58 versus 59 at h24; the common 59-predictor budget was compared with the full model at h10, h13, and h24. Complete outcomes are reported in Section 3.8.

Reproducibility (supports Section 2.12)

Reproducibility was supported by frozen protocol documents, source and output SHA-256 hashes, machine-readable JSON assertions, exact feature lists and family maps, saved Optuna studies and selected parameters, row-level Parquet predictions, preprocessing-parameter tables, fold-boundary tables, complete h1–h24 metrics, and exact SHAP artifacts. The active implementation used Python 3.13.9, pandas 2.3.3, NumPy 2.1.3, CatBoost 1.2.10, Optuna 4.9.0, scikit-learn 1.8.0, SHAP 0.52.0, and PyArrow 21.0.0. Randomized procedures used seed 42 unless a deterministic repeat schedule was explicitly specified. Historical analyses based on an incorrect geographic merge, capped held-out targets, incomplete winter coverage, or superseded event selections were isolated from the active publication package and were not used in this manuscript.

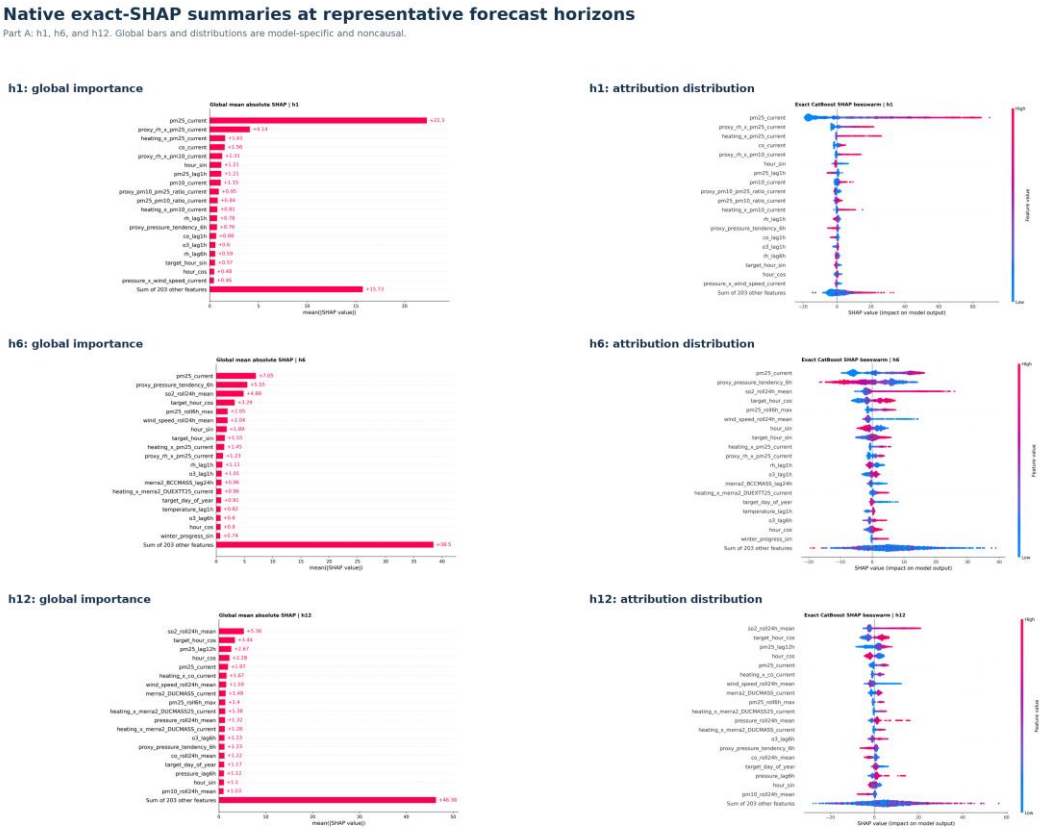
Supplementary figures



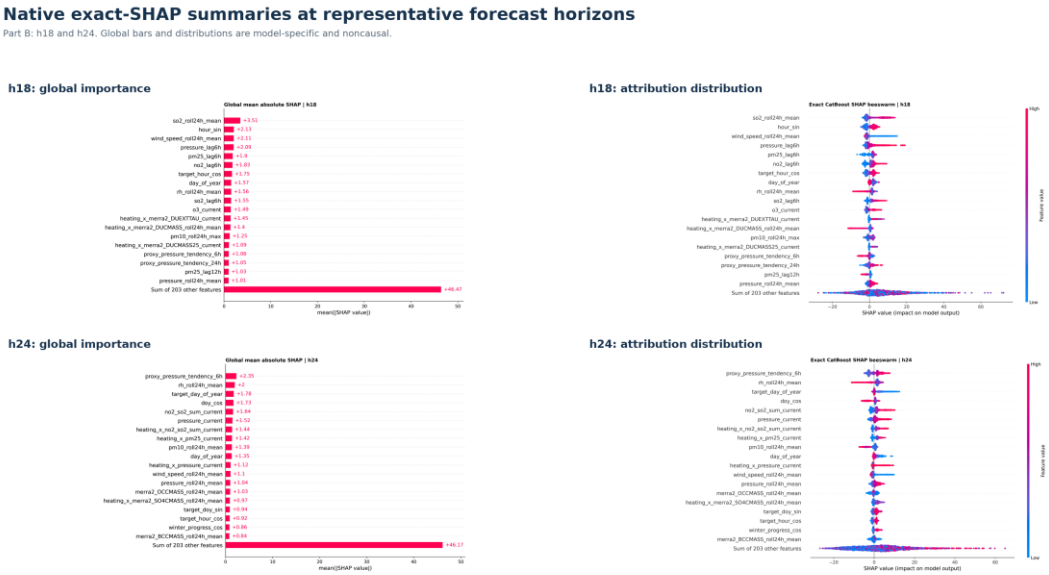
Supplementary Figure S1. Exact feature-level SHAP evolution across h1–h24 for the primary 222-predictor full models (with MERRA-2). Individual-feature rankings are presented as model attribution rather than causal or independently incremental effects; family-level summaries remain the primary interpretation because correlated predictors may redistribute attribution.



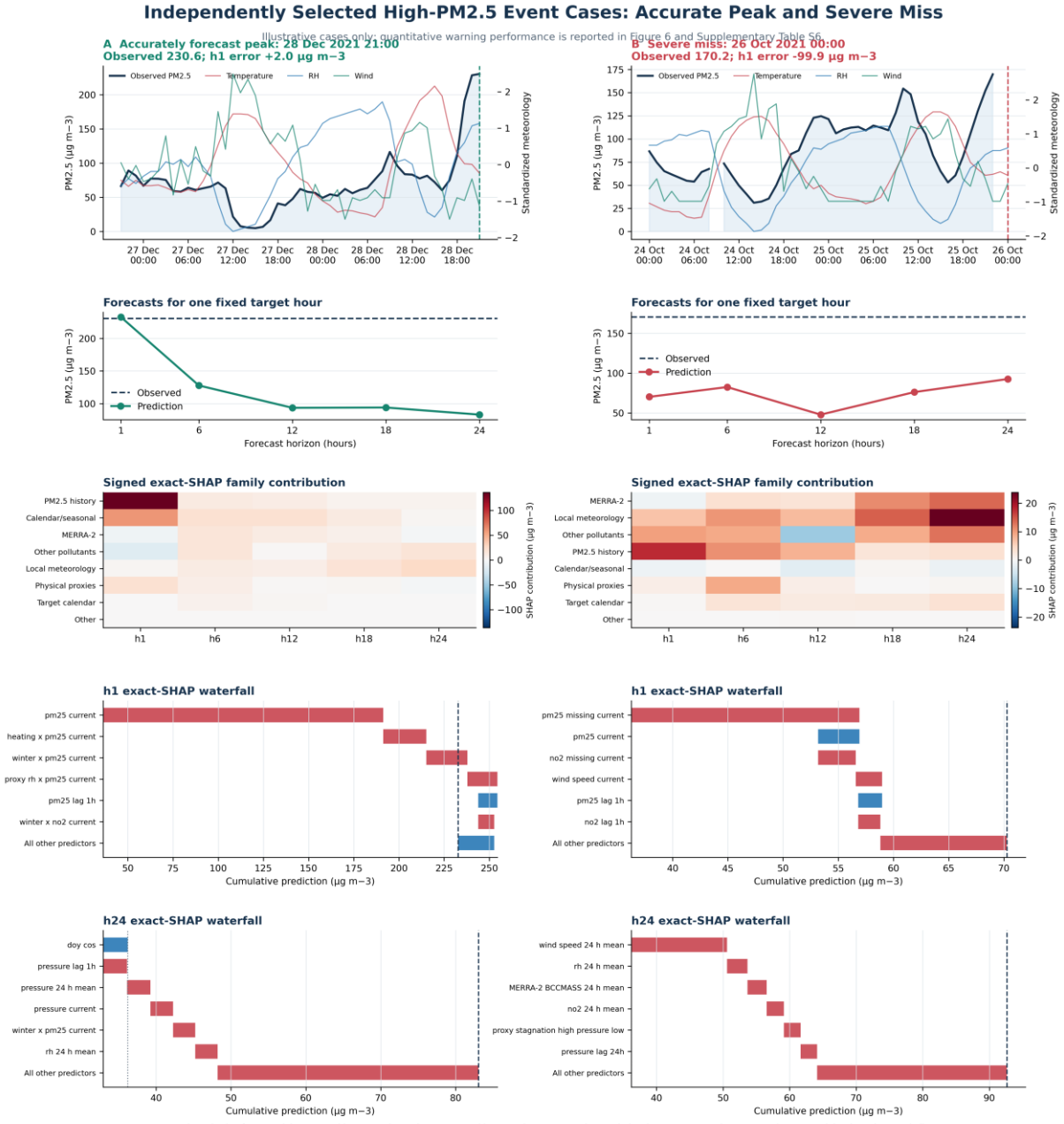
Supplementary Figure S2. Prespecified validation-only feature-reduction acceptance decision. Horizon-specific reduced model minus full model R^2 and relative RMSE and MAE changes are compared with the frozen acceptance limits. For R^2, negative values mean the reduced model was below the full model. The individual R^2, RMSE, and MAE criteria each passed at 19 of 24 horizons, and 17 horizons passed all three jointly. The reduced version therefore remained supporting validation evidence, and the full 222-predictor model remained primary. Error bars show paired 95% moving-block-bootstrap intervals for reduced model minus full model R^2.



Supplementary Figure S3A. Native exact-SHAP global-importance bars and attribution-distribution plots for h1, h6, and h12. The enlarged panels preserve the native feature-level summaries while improving print readability. These representative summaries complement the family-level main-text analysis; SHAP remains model-specific and noncausal.



Supplementary Figure S3B. Native exact-SHAP global-importance bars and attribution-distribution plots for h18 and h24. The enlarged panels complete the representative horizon sequence shown in Supplementary Figure S3A. SHAP remains model-specific and noncausal.


Independently Selected High-PM2.5 Event Cases: Accurate Peak and Severe Miss
Illustrative cases only; quantitative warning performance is reported in Figure 6 and Supplementary Table S6.
A Accurately forecast peak: 28 Dec 2021 21:00
Observed 230.6; h1 error +2.0 μg m−3
B Severe miss: 26 Oct 2021 00:00
Observed 170.2; h1 error -99.9 μg m−3
Observed PM2.5
Temperature
RH
Wind
PM2.5 (μg m−3)
Standardized meteorology
Forecasts for one fixed target hour
Observed
Prediction
Forecast horizon (hours)
Signed exact-SHAP family contribution
PM2.5 history
Calendar/seasonal
MERRA-2
Other pollutants
Local meteorology
Physical proxies
Target calendar
Other
SHAP contribution (μg m−3)
h1 exact-SHAP waterfall
pm25 current
heating x pm25 current
winter x pm25 current
proxy rh x pm25 current
pm25 lag 1h
winter x no2 current
All other predictors
Cumulative prediction (μg m−3)
pm25 missing current
no2 missing current
wind speed current
no2 lag 1h
h24 exact-SHAP waterfall
doy cos
pressure lag 1h
pressure 24 h mean
pressure current
winter x pm25 current
rh 24 h mean
wind speed 24 h mean
MERRA-2 BCCMASS 24 h mean
no2 24 h mean
proxy stagnation high pressure low
pressure lag 24h
Cases were selected using frozen training-P95 and h1-error rules and are separated by 1,533 h. Exact SHAP is associational; MERRA-2 uses the retrospective area-weighted CST hour-end alignment.

Supplementary Figure S4. Independently selected high-PM2.5 event diagnostics for an accurately forecast peak and a severe peak with large forecast error. For each fixed target hour, panels show the preceding 48 h pollution and standardized meteorological context, forecasts at h1/h6/h12/h18/h24, signed family-level exact-SHAP contributions, and h1/h24 exact-SHAP waterfalls. The cases were selected using frozen training-P95 and h1-error rules and are illustrative only; Figure 7 and Supplementary Table S6 provide the quantitative warning evaluation.

Supplementary Table S2. Complete locked-test metrics across h1–h24

All values use the archived primary prediction sets and uncapped observed targets. Persistence has fewer available rows when forecast-origin PM2.5 is missing.

Model	Horizon	R^2	RMSE (µg m−3)	MAE (µg m−3)	Bias (µg m−3)	n	Predictors
Full model (with MERRA-2)	h1	0.9488	7.019	4.157	0.415	3373	222.0
Full model (with MERRA-2)	h2	0.8486	12.075	7.308	0.926	3373	222.0
Full model (with MERRA-2)	h3	0.7736	14.766	9.024	1.262	3373	222.0
Full model (with MERRA-2)	h4	0.7071	16.793	10.532	1.362	3373	222.0
Full model (with MERRA-2)	h5	0.6390	18.643	11.711	1.762	3373	222.0
Full model (with MERRA-2)	h6	0.5876	19.928	12.679	1.791	3373	222.0
Full model (with MERRA-2)	h7	0.5446	20.941	13.354	1.608	3373	222.0
Full model (with MERRA-2)	h8	0.4694	22.602	15.225	3.354	3373	222.0
Full model (with MERRA-2)	h9	0.4722	22.543	14.870	2.148	3373	222.0
Full model (with MERRA-2)	h10	0.4361	23.301	15.536	2.149	3373	222.0
Full model (with MERRA-2)	h11	0.4004	24.029	16.059	1.856	3373	222.0

Model	Horizon	R^2	RMSE (µg m−3)	MAE (µg m−3)	Bias (µg m−3)	n	Predictors
Full model (with MERRA-2)	h12	0.3809	24.416	16.529	2.350	3373	222.0
Full model (with MERRA-2)	h13	0.3613	24.799	16.823	1.774	3373	222.0
Full model (with MERRA-2)	h14	0.3737	24.557	16.917	2.198	3373	222.0
Full model (with MERRA-2)	h15	0.3252	25.490	17.316	0.912	3373	222.0
Full model (with MERRA-2)	h16	0.3326	25.350	17.522	1.076	3373	222.0
Full model (with MERRA-2)	h17	0.3541	24.939	17.414	1.982	3373	222.0
Full model (with MERRA-2)	h18	0.3388	25.233	17.781	1.900	3373	222.0
Full model (with MERRA-2)	h19	0.3065	25.841	18.255	2.484	3373	222.0
Full model (with MERRA-2)	h20	0.2774	26.377	18.327	1.773	3373	222.0
Full model (with MERRA-2)	h21	0.3146	25.690	18.041	2.608	3373	222.0
Full model (with MERRA-2)	h22	0.2894	26.157	18.215	2.045	3373	222.0
Full model (with MERRA-2)	h23	0.2655	26.594	18.576	2.481	3373	222.0
Full model (with MERRA-2)	h24	0.2044	27.677	19.284	2.086	3373	222.0

Model	Horizon	R^2	RMSE (µg m−3)	MAE (µg m−3)	Bias (µg m−3)	n	Predictors
Local-only model	h1	0.9467	7.167	4.133	0.310	3373	149.0
Local-only model	h2	0.8653	11.389	6.840	0.214	3373	149.0
Local-only model	h3	0.7734	14.773	8.909	0.449	3373	149.0
Local-only model	h4	0.7024	16.929	10.275	0.155	3373	149.0
Local-only model	h5	0.6356	18.732	11.738	0.774	3373	149.0
Local-only model	h6	0.5871	19.939	12.577	0.529	3373	149.0
Local-only model	h7	0.5351	21.157	13.356	0.129	3373	149.0
Local-only model	h8	0.4678	22.636	14.617	1.125	3373	149.0
Local-only model	h9	0.4027	23.982	15.638	1.155	3373	149.0
Local-only model	h10	0.3059	25.853	17.108	0.840	3373	149.0
Local-only model	h11	0.3593	24.838	16.382	0.041	3373	149.0
Local-only model	h12	0.3193	25.601	16.860	0.192	3373	149.0
Local-only model	h13	0.2989	25.982	17.159	-0.340	3373	149.0

Model	Horizon	R^2	RMSE (µg m−3)	MAE (µg m−3)	Bias (µg m−3)	n	Predictors
Local-only model	h14	0.3043	25.881	17.536	0.192	3373	149.0
Local-only model	h15	0.2743	26.435	17.851	-0.197	3373	149.0
Local-only model	h16	0.2236	27.342	18.296	-0.765	3373	149.0
Local-only model	h17	0.2698	26.516	17.969	-0.580	3373	149.0
Local-only model	h18	0.2490	26.891	18.282	-0.239	3373	149.0
Local-only model	h19	0.2320	27.193	18.745	0.188	3373	149.0
Local-only model	h20	0.2401	27.049	18.381	-0.504	3373	149.0
Local-only model	h21	0.2174	27.452	18.970	-0.044	3373	149.0
Local-only model	h22	0.2300	27.230	18.726	-0.646	3373	149.0
Local-only model	h23	0.2498	26.876	18.535	-0.486	3373	149.0
Local-only model	h24	0.2102	27.577	19.059	-0.102	3373	149.0
Month–hour climatology	h1	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h2	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h3	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h4	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h5	0.1550	28.524	20.063	1.155	3373	

Model	Horizon	R^2	RMSE (µg m−3)	MAE (µg m−3)	Bias (µg m−3)	n	Predictors
Month–hour climatology	h6	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h7	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h8	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h9	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h10	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h11	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h12	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h13	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h14	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h15	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h16	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h17	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h18	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h19	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h20	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h21	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h22	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h23	0.1550	28.524	20.063	1.155	3373	
Month–hour climatology	h24	0.1550	28.524	20.063	1.155	3373	
Persistence	h1	0.9169	8.935	4.935	-0.016	3329	
Persistence	h2	0.7535	15.388	8.628	-0.037	3325	
Persistence	h3	0.5738	20.226	11.564	-0.062	3327	
Persistence	h4	0.4082	23.838	13.902	-0.077	3328	
Persistence	h5	0.2714	26.470	15.780	-0.104	3324	
Persistence	h6	0.1646	28.356	17.262	-0.117	3321	
Persistence	h7	0.0904	29.606	18.242	-0.154	3320	
Persistence	h8	0.0426	30.372	18.943	-0.180	3321	
Persistence	h9	0.0201	30.796	19.393	-0.264	3319	
Persistence	h10	0.0034	31.105	19.880	-0.343	3319	
Persistence	h11	-0.0225	31.506	20.425	-0.355	3317	
Persistence	h12	-0.0618	32.094	20.999	-0.369	3315	
Persistence	h13	-0.1181	32.921	21.729	-0.382	3315	

Model	Horizon	R^2	RMSE (µg m−3)	MAE (µg m−3)	Bias (µg m−3)	n	Predictors
Persistence	h14	-0.1869	33.907	22.466	-0.421	3316	
Persistence	h15	-0.2571	34.889	23.149	-0.444	3316	
Persistence	h16	-0.3157	35.722	23.658	-0.486	3311	
Persistence	h17	-0.3639	36.357	24.021	-0.446	3311	
Persistence	h18	-0.3690	36.394	24.022	-0.477	3312	
Persistence	h19	-0.3547	36.159	23.805	-0.466	3309	
Persistence	h20	-0.3219	35.718	23.409	-0.479	3308	
Persistence	h21	-0.2782	35.117	22.948	-0.466	3306	
Persistence	h22	-0.2291	34.432	22.418	-0.467	3307	
Persistence	h23	-0.1938	33.945	22.102	-0.456	3305	
Persistence	h24	-0.1930	33.788	22.163	-0.424	3306	

Supplementary Table S3. Selected CatBoost hyperparameters by model and horizon

The lowest-validation-RMSE completed trial was selected within each 50-trial seeded study. Each search used 20 startup trials and then the adaptive TPE phase; the locked test was not used for selection.

Model	Horizon	Iterations	Depth	Learning rate	L2 leaf reg.	Random strength	Bagging temperature	Border count	Seed
Local-only model	h1	374	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h1	694	4	0.06788	1.7313	0.3253	4.7444	248	42
Local-only model	h2	274	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h2	292	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h3	329	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h3	254	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h4	258	7	0.08714	1.0685	4.8495	4.1622	104	42
Full model (with MERRA-2)	h4	300	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h5	435	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h5	220	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h6	420	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h6	319	8	0.10294	20.5761	4.4741	2.9895	240	42
Local-only model	h7	857	5	0.01912	9.0473	2.2008	0.6102	158	42
Full model (with MERRA-2)	h7	190	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h8	316	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h8	198	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h9	206	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h9	508	5	0.05778	1.5740	4.0110	0.3728	252	42
Local-only model	h10	839	5	0.01912	9.0473	2.2008	0.6102	158	42
Full model (with MERRA-2)	h10	486	5	0.05778	1.5740	4.0110	0.3728	252	42
Local-only model	h11	160	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h11	198	7	0.08714	1.0685	4.8495	4.1622	104	42
Local-only model	h12	121	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h12	183	8	0.10294	20.5761	4.4741	2.9895	240	42
Local-only model	h13	131	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h13	181	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h14	134	7	0.08714	1.0685	4.8495	4.1622	104	42
Full model (with MERRA-2)	h14	233	7	0.08714	1.0685	4.8495	4.1622	104	42
Local-only model	h15	310	5	0.05778	1.5740	4.0110	0.3728	252	42
Full model (with MERRA-2)	h15	229	7	0.08714	1.0685	4.8495	4.1622	104	42
Local-only model	h16	122	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h16	376	8	0.10294	20.5761	4.4741	2.9895	240	42
Local-only model	h17	163	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h17	193	8	0.10294	20.5761	4.4741	2.9895	240	42
Local-only model	h18	186	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h18	163	8	0.09248	6.8689	0.7801	0.7800	75	42
Local-only model	h19	254	8	0.10294	20.5761	4.4741	2.9895	240	42

Model	Horizon	Iterations	Depth	Learning rate	L2 leaf reg.	Random strength	Bagging temperature	Border count	Seed
Full model (with MERRA-2)	h19	287	8	0.10294	20.5761	4.4741	2.9895	240	42
Local-only model	h20	465	4	0.06788	1.7313	0.3253	4.7444	248	42
Full model (with MERRA-2)	h20	517	5	0.05778	1.5740	4.0110	0.3728	252	42
Local-only model	h21	167	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h21	257	7	0.08714	1.0685	4.8495	4.1622	104	42
Local-only model	h22	218	8	0.10294	20.5761	4.4741	2.9895	240	42
Full model (with MERRA-2)	h22	513	5	0.05778	1.5740	4.0110	0.3728	252	42
Local-only model	h23	442	5	0.05778	1.5740	4.0110	0.3728	252	42
Full model (with MERRA-2)	h23	268	8	0.10294	20.5761	4.4741	2.9895	240	42
Local-only model	h24	131	8	0.09248	6.8689	0.7801	0.7800	75	42
Full model (with MERRA-2)	h24	517	5	0.05778	1.5740	4.0110	0.3728	252	42

Supplementary Table S4. Complete predictor names, families, and availability rules

The 222-predictor contract contains only forecast-origin information or deterministic target-calendar fields known at issuance. Draft 3 selected the primary models with 50 validation-only Optuna trials per search.

Predictor	Family	Availability rule
hour_sin	Calendar/seasonal timing	forecast origin t or earlier
hour_cos	Calendar/seasonal timing	forecast origin t or earlier
doy_sin	Calendar/seasonal timing	forecast origin t or earlier
doy_cos	Calendar/seasonal timing	forecast origin t or earlier
month	Calendar/seasonal timing	forecast origin t or earlier
season	Calendar/seasonal timing	forecast origin t or earlier
pm25_current	PM2.5 history	forecast origin t or earlier
pm25_lag1h	PM2.5 history	forecast origin t or earlier
pm25_lag3h	PM2.5 history	forecast origin t or earlier
pm25_lag6h	PM2.5 history	forecast origin t or earlier
pm25_lag12h	PM2.5 history	forecast origin t or earlier
pm25_lag24h	PM2.5 history	forecast origin t or earlier
pm25_missing_current	PM2.5 history	forecast origin t or earlier
pm25_roll6h_mean	PM2.5 history	forecast origin t or earlier
pm25_roll6h_max	PM2.5 history	forecast origin t or earlier
pm25_roll6h_std	PM2.5 history	forecast origin t or earlier
pm25_roll24h_mean	PM2.5 history	forecast origin t or earlier
pm25_roll24h_max	PM2.5 history	forecast origin t or earlier
pm25_roll24h_std	PM2.5 history	forecast origin t or earlier
pm10_current	Other pollutant history	forecast origin t or earlier
pm10_lag1h	Other pollutant history	forecast origin t or earlier
pm10_lag3h	Other pollutant history	forecast origin t or earlier
pm10_lag6h	Other pollutant history	forecast origin t or earlier
pm10_lag12h	Other pollutant history	forecast origin t or earlier
pm10_lag24h	Other pollutant history	forecast origin t or earlier
pm10_missing_current	Other pollutant history	forecast origin t or earlier
pm10_roll6h_mean	Other pollutant history	forecast origin t or earlier
pm10_roll6h_max	Other pollutant history	forecast origin t or earlier
pm10_roll24h_mean	Other pollutant history	forecast origin t or earlier
pm10_roll24h_max	Other pollutant history	forecast origin t or earlier
no2_current	Other pollutant history	forecast origin t or earlier
no2_lag1h	Other pollutant history	forecast origin t or earlier
no2_lag6h	Other pollutant history	forecast origin t or earlier
no2_lag24h	Other pollutant history	forecast origin t or earlier
no2_missing_current	Other pollutant history	forecast origin t or earlier
no2_roll24h_mean	Other pollutant history	forecast origin t or earlier
so2_current	Other pollutant history	forecast origin t or earlier

Predictor	Family	Availability rule
so2_lag1h	Other pollutant history	forecast origin t or earlier
so2_lag6h	Other pollutant history	forecast origin t or earlier
so2_lag24h	Other pollutant history	forecast origin t or earlier
so2_missing_current	Other pollutant history	forecast origin t or earlier
so2_roll24h_mean	Other pollutant history	forecast origin t or earlier
co_current	Other pollutant history	forecast origin t or earlier
co_lag1h	Other pollutant history	forecast origin t or earlier
co_lag6h	Other pollutant history	forecast origin t or earlier
co_lag24h	Other pollutant history	forecast origin t or earlier
co_missing_current	Other pollutant history	forecast origin t or earlier
co_roll24h_mean	Other pollutant history	forecast origin t or earlier
temperature_current	Local meteorology	forecast origin t or earlier
temperature_lag1h	Local meteorology	forecast origin t or earlier
temperature_lag6h	Local meteorology	forecast origin t or earlier
temperature_lag24h	Local meteorology	forecast origin t or earlier
temperature_missing_current	Local meteorology	forecast origin t or earlier
temperature_roll24h_mean	Local meteorology	forecast origin t or earlier
o3_current	Other pollutant history	forecast origin t or earlier
o3_lag1h	Other pollutant history	forecast origin t or earlier
o3_lag6h	Other pollutant history	forecast origin t or earlier
o3_lag24h	Other pollutant history	forecast origin t or earlier
o3_missing_current	Other pollutant history	forecast origin t or earlier
o3_roll24h_mean	Other pollutant history	forecast origin t or earlier
rh_current	Local meteorology	forecast origin t or earlier
rh_lag1h	Local meteorology	forecast origin t or earlier
rh_lag6h	Local meteorology	forecast origin t or earlier
rh_lag24h	Local meteorology	forecast origin t or earlier
rh_missing_current	Local meteorology	forecast origin t or earlier
rh_roll24h_mean	Local meteorology	forecast origin t or earlier
pressure_current	Local meteorology	forecast origin t or earlier
pressure_lag1h	Local meteorology	forecast origin t or earlier
pressure_lag6h	Local meteorology	forecast origin t or earlier
pressure_lag24h	Local meteorology	forecast origin t or earlier
pressure_missing_current	Local meteorology	forecast origin t or earlier
pressure_roll24h_mean	Local meteorology	forecast origin t or earlier
wind_speed_current	Local meteorology	forecast origin t or earlier
wind_speed_lag1h	Local meteorology	forecast origin t or earlier
wind_speed_lag6h	Local meteorology	forecast origin t or earlier
wind_speed_lag24h	Local meteorology	forecast origin t or earlier
wind_speed_missing_current	Local meteorology	forecast origin t or earlier
wind_speed_roll24h_mean	Local meteorology	forecast origin t or earlier
pm25_pm10_ratio_current	PM2.5 history	forecast origin t or earlier
pm10_minus_pm25_current	Other pollutant history	forecast origin t or earlier
no2_so2_sum_current	Other pollutant history	forecast origin t or earlier
temp_x_rh_current	Local meteorology	forecast origin t or earlier

Predictor	Family	Availability rule
pressure_x_wind_speed_current	Local meteorology	forecast origin t or earlier
day_of_year	Calendar/seasonal timing	forecast origin t or earlier
month_sin	Calendar/seasonal timing	forecast origin t or earlier
month_cos	Calendar/seasonal timing	forecast origin t or earlier
is_winter	Calendar/seasonal timing	forecast origin t or earlier
is_heating_season	Calendar/seasonal timing	forecast origin t or earlier
is_cold_season	Other	forecast origin t or earlier
winter_progress_sin	Calendar/seasonal timing	forecast origin t or earlier
winter_progress_cos	Calendar/seasonal timing	forecast origin t or earlier
winter_x_pm25_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_pm25_current	Calendar/seasonal timing	forecast origin t or earlier
winter_x_pm25_roll24h_mean	Calendar/seasonal timing	forecast origin t or earlier
heating_x_pm25_roll24h_mean	Calendar/seasonal timing	forecast origin t or earlier
winter_x_pm10_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_pm10_current	Calendar/seasonal timing	forecast origin t or earlier
winter_x_no2_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_no2_current	Calendar/seasonal timing	forecast origin t or earlier
winter_x_so2_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_so2_current	Calendar/seasonal timing	forecast origin t or earlier
winter_x_co_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_co_current	Calendar/seasonal timing	forecast origin t or earlier
winter_x_temperature_current	Local meteorology	forecast origin t or earlier
heating_x_temperature_current	Local meteorology	forecast origin t or earlier
winter_x_rh_current	Local meteorology	forecast origin t or earlier
heating_x_rh_current	Local meteorology	forecast origin t or earlier
winter_x_pressure_current	Local meteorology	forecast origin t or earlier
heating_x_pressure_current	Local meteorology	forecast origin t or earlier
winter_x_wind_speed_current	Local meteorology	forecast origin t or earlier
heating_x_wind_speed_current	Local meteorology	forecast origin t or earlier
winter_x_o3_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_o3_current	Calendar/seasonal timing	forecast origin t or earlier
winter_x_temp_x_rh_current	Local meteorology	forecast origin t or earlier
heating_x_temp_x_rh_current	Local meteorology	forecast origin t or earlier
winter_x_no2_so2_sum_current	Calendar/seasonal timing	forecast origin t or earlier
heating_x_no2_so2_sum_current	Calendar/seasonal timing	forecast origin t or earlier
proxy_wind_dir_sin_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_wind_dir_cos_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_wind_calm	Engineered local-condition indicators	forecast origin t or earlier
proxy_wind_low	Engineered local-condition indicators	forecast origin t or earlier
proxy_wind_moderate	Engineered local-condition indicators	forecast origin t or earlier
proxy_wind_high	Engineered local-condition indicators	forecast origin t or earlier
proxy_wind_speed_tendency_6h	Engineered local-condition indicators	forecast origin t or earlier
proxy_ventilation_wind_only	Engineered local-condition indicators	forecast origin t or earlier
proxy_ventilation_low_flag	Engineered local-condition indicators	forecast origin t or earlier
proxy_stagnation_high_pressure_low_wind	Engineered local-condition indicators	forecast origin t or earlier

Predictor	Family	Availability rule
proxy_pressure_tendency_6h	Engineered local-condition indicators	forecast origin t or earlier
proxy_pressure_tendency_24h	Engineered local-condition indicators	forecast origin t or earlier
proxy_rh_x_pm10_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_rh_x_pm25_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_rh_x_temperature_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_precip_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_precip_any_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_precip_roll6h_sum	Engineered local-condition indicators	forecast origin t or earlier
proxy_precip_roll24h_sum	Engineered local-condition indicators	forecast origin t or earlier
proxy_recent_wet_suppression_flag	Engineered local-condition indicators	forecast origin t or earlier
proxy_pm10_pm25_ratio_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_pmcoarse_current	Engineered local-condition indicators	forecast origin t or earlier
proxy_pmcoarse_floor0_current	Engineered local-condition indicators	forecast origin t or earlier
merra2_origin_available	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS25_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS25_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS25_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS25_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUCMASS25_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTTAU_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTTAU_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTTAU_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTTAU_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTTAU_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTT25_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTT25_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTT25_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTT25_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_DUEXTT25_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_TOTEXTTAU_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_TOTEXTTAU_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_TOTEXTTAU_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_TOTEXTTAU_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_TOTEXTTAU_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_SO4CMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_SO4CMASS_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_SO4CMASS_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_SO4CMASS_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_SO4CMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_BCCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier

Predictor	Family	Availability rule
merra2_BCCMASS_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_BCCMASS_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_BCCMASS_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_BCCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_OCCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_OCCMASS_lag1h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_OCCMASS_lag6h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_OCCMASS_lag24h	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
merra2_OCCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUCMASS25_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUCMASS25_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUCMASS25_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUCMASS25_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUEXTTAU_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUEXTTAU_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUEXTTAU_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUEXTTAU_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUEXTT25_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUEXTT25_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_DUEXTT25_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_DUEXTT25_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_TOTEXTTAU_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_TOTEXTTAU_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_TOTEXTTAU_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_TOTEXTTAU_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_SO4CMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_SO4CMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_SO4CMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_SO4CMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_BCCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_BCCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_BCCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_BCCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_OCCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_OCCMASS_current	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
winter_x_merra2_OCCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
heating_x_merra2_OCCMASS_roll24h_mean	MERRA-2 aerosol/reanalysis	forecast origin t or earlier
target_hour_sin	Known target-calendar timing	known deterministically at forecast origin
target_hour_cos	Known target-calendar timing	known deterministically at forecast origin
target_doy_sin	Known target-calendar timing	known deterministically at forecast origin
target_doy_cos	Known target-calendar timing	known deterministically at forecast origin

Predictor	Family	Availability rule
target_day_of_year	Known target-calendar timing	known deterministically at forecast origin
target_month_sin	Known target-calendar timing	known deterministically at forecast origin
target_month_cos	Known target-calendar timing	known deterministically at forecast origin
target_is_winter	Known target-calendar timing	known deterministically at forecast origin
target_is_heating_season	Known target-calendar timing	known deterministically at forecast origin

Supplementary Table S5. Seasonal h1–h24 performance and paired R²-difference intervals

Spring, summer, and autumn use their rolling-origin folds. Composite winter pools January–February and December 2021 predictions from two chronologically ordered folds; bootstrap blocks never cross the intervening gap.

Season	Horizon	Local R²	MERRA-2 R²	ΔR²	95% CI lower	95% CI upper	MERRA-2 RMSE (µg m−3)	MERRA-2 MAE (µg m−3)	n
Spring	h1	0.9315	0.9307	-0.0008	-0.0036	0.0017	9.722	5.805	2208
Spring	h2	0.8295	0.8263	-0.0031	-0.0080	0.0016	15.386	9.328	2208
Spring	h3	0.7336	0.7408	0.0071	-0.0064	0.0181	18.798	11.723	2208
Spring	h4	0.6620	0.6683	0.0063	-0.0163	0.0251	21.265	13.668	2208
Spring	h5	0.6040	0.6093	0.0053	-0.0213	0.0290	23.077	15.226	2208
Spring	h6	0.5535	0.5492	-0.0043	-0.0349	0.0287	24.788	16.574	2208
Spring	h7	0.4907	0.5073	0.0166	-0.0359	0.0709	25.915	17.170	2208
Spring	h8	0.4839	0.4756	-0.0083	-0.0454	0.0218	26.736	18.009	2208
Spring	h9	0.4478	0.4460	-0.0018	-0.0452	0.0448	27.481	18.661	2208
Spring	h10	0.4069	0.3852	-0.0217	-0.0802	0.0359	28.949	19.647	2208
Spring	h11	0.3814	0.3684	-0.0130	-0.0739	0.0446	29.342	19.823	2208
Spring	h12	0.3776	0.3649	-0.0126	-0.0817	0.0503	29.422	19.854	2208
Spring	h13	0.3646	0.3554	-0.0092	-0.0971	0.0686	29.641	20.032	2208
Spring	h14	0.3745	0.3431	-0.0314	-0.1090	0.0368	29.923	20.342	2208
Spring	h15	0.3323	0.3409	0.0086	-0.0581	0.0656	29.974	19.917	2208
Spring	h16	0.3126	0.3021	-0.0105	-0.0811	0.0567	30.844	20.682	2208
Spring	h17	0.3165	0.3099	-0.0066	-0.0990	0.0774	30.672	20.735	2208
Spring	h18	0.2986	0.2924	-0.0063	-0.0839	0.0714	31.058	20.564	2208
Spring	h19	0.2871	0.2800	-0.0071	-0.0894	0.0675	31.328	20.745	2208
Spring	h20	0.2643	0.2995	0.0352	-0.0576	0.1073	30.900	21.014	2208
Spring	h21	0.2649	0.3051	0.0402	-0.0304	0.1086	30.777	20.524	2208
Spring	h22	0.2659	0.3027	0.0368	-0.0349	0.1089	30.830	20.462	2208
Spring	h23	0.2875	0.3188	0.0314	-0.0235	0.0837	30.472	20.767	2208
Spring	h24	0.2556	0.3071	0.0516	-0.0040	0.1098	30.732	20.599	2208
Summer	h1	0.9060	0.9067	0.0007	-0.0024	0.0034	4.264	3.008	2091
Summer	h2	0.7767	0.7759	-0.0008	-0.0080	0.0053	6.609	4.659	2091
Summer	h3	0.6651	0.6702	0.0052	-0.0090	0.0183	8.018	5.757	2091
Summer	h4	0.5839	0.5810	-0.0029	-0.0207	0.0125	9.038	6.562	2091
Summer	h5	0.5087	0.5154	0.0068	-0.0141	0.0278	9.718	7.071	2091
Summer	h6	0.4642	0.4670	0.0028	-0.0246	0.0298	10.193	7.368	2091
Summer	h7	0.4188	0.4243	0.0055	-0.0244	0.0316	10.593	7.696	2091
Summer	h8	0.3963	0.3966	0.0003	-0.0455	0.0486	10.845	7.831	2091
Summer	h9	0.3690	0.3857	0.0167	-0.0354	0.0575	10.943	7.941	2091
Summer	h10	0.3707	0.3864	0.0157	-0.0366	0.0656	10.936	8.037	2091
Summer	h11	0.3461	0.3594	0.0133	-0.0287	0.0544	11.174	8.129	2091
Summer	h12	0.2961	0.3536	0.0575	0.0110	0.0988	11.225	8.161	2091
Summer	h13	0.2721	0.3129	0.0407	0.0071	0.0729	11.573	8.489	2091

Season	Horizon	Local R^2	MERRA-2 R^2	ΔR^2	95% CI lower	95% CI upper	MERRA-2 RMSE (μg m−3)	MERRA-2 MAE (μg m−3)	n
Summer	h14	0.2485	0.2903	0.0418	0.0033	0.0840	11.761	8.722	2091
Summer	h15	0.2496	0.2668	0.0172	-0.0198	0.0529	11.955	8.851	2091
Summer	h16	0.2278	0.2602	0.0324	-0.0118	0.0777	12.008	8.854	2091
Summer	h17	0.1976	0.2381	0.0405	0.0075	0.0729	12.186	9.000	2091
Summer	h18	0.1965	0.2465	0.0500	0.0143	0.0891	12.119	8.884	2091
Summer	h19	0.2206	0.2438	0.0232	-0.0092	0.0571	12.141	9.014	2091
Summer	h20	0.2116	0.2436	0.0320	-0.0044	0.0686	12.142	9.025	2091
Summer	h21	0.2297	0.2571	0.0274	-0.0233	0.0659	12.033	9.021	2091
Summer	h22	0.2110	0.2449	0.0339	-0.0127	0.0806	12.132	9.057	2091
Summer	h23	0.2006	0.2238	0.0232	-0.0371	0.0710	12.300	9.153	2091
Summer	h24	0.2166	0.2200	0.0034	-0.0583	0.0615	12.330	9.330	2091
Autumn	h1	0.9490	0.9470	-0.0020	-0.0041	0.0002	7.529	4.214	2163
Autumn	h2	0.8691	0.8708	0.0017	-0.0068	0.0107	11.757	6.907	2163
Autumn	h3	0.7911	0.7877	-0.0034	-0.0159	0.0086	15.071	9.025	2163
Autumn	h4	0.7179	0.7250	0.0071	-0.0121	0.0241	17.152	10.729	2163
Autumn	h5	0.6656	0.6671	0.0014	-0.0316	0.0296	18.873	11.928	2163
Autumn	h6	0.6224	0.6289	0.0066	-0.0260	0.0316	19.924	12.647	2163
Autumn	h7	0.5799	0.5729	-0.0070	-0.0498	0.0237	21.375	13.585	2163
Autumn	h8	0.5251	0.5293	0.0042	-0.0488	0.0409	22.440	14.453	2163
Autumn	h9	0.4641	0.4949	0.0308	-0.0457	0.0843	23.247	14.932	2163
Autumn	h10	0.4246	0.4769	0.0523	-0.0033	0.0971	23.656	15.384	2163
Autumn	h11	0.4103	0.4620	0.0518	-0.0373	0.1125	23.990	15.959	2163
Autumn	h12	0.4142	0.4384	0.0242	-0.0732	0.1092	24.512	16.484	2163
Autumn	h13	0.3675	0.4089	0.0414	-0.0436	0.1048	25.147	16.849	2163
Autumn	h14	0.3221	0.3966	0.0744	-0.0195	0.1500	25.407	17.200	2163
Autumn	h15	0.3024	0.3890	0.0866	-0.0139	0.1810	25.567	17.158	2163
Autumn	h16	0.2949	0.3879	0.0930	0.0157	0.1686	25.589	17.006	2163
Autumn	h17	0.2770	0.3888	0.1118	0.0124	0.2024	25.571	17.534	2163
Autumn	h18	0.2596	0.3695	0.1098	0.0149	0.2004	25.972	17.604	2163
Autumn	h19	0.2532	0.3515	0.0983	0.0070	0.1719	26.340	17.725	2163
Autumn	h20	0.2813	0.3551	0.0738	0.0116	0.1232	26.265	17.655	2163
Autumn	h21	0.2640	0.3250	0.0610	-0.0126	0.1146	26.872	17.684	2163
Autumn	h22	0.2583	0.3381	0.0799	0.0178	0.1416	26.609	17.720	2163
Autumn	h23	0.2786	0.3425	0.0639	-0.0295	0.1401	26.521	17.971	2163
Autumn	h24	0.2396	0.3334	0.0938	0.0248	0.1569	26.705	18.118	2163
Winter	h1	0.9433	0.9292	-0.0141	-0.0241	0.0005	12.980	7.220	2142
Winter	h2	0.8429	0.8378	-0.0051	-0.0174	0.0110	19.646	11.797	2142
Winter	h3	0.7341	0.7286	-0.0055	-0.0207	0.0101	25.415	15.436	2142
Winter	h4	0.6354	0.6264	-0.0090	-0.0272	0.0120	29.821	18.841	2142
Winter	h5	0.5367	0.5277	-0.0090	-0.0465	0.0256	33.529	21.589	2142
Winter	h6	0.4428	0.4497	0.0069	-0.0308	0.0410	36.192	23.890	2142
Winter	h7	0.3949	0.4079	0.0130	-0.0209	0.0422	37.543	25.179	2142
Winter	h8	0.3482	0.3595	0.0113	-0.0336	0.0531	39.045	26.660	2142
Winter	h9	0.3435	0.3475	0.0040	-0.0429	0.0495	39.409	27.009	2142
Winter	h10	0.3030	0.3116	0.0086	-0.0409	0.0595	40.478	27.877	2142

Season	Horizon	Local R^2	MERRA-2 R^2	ΔR^2	95% CI lower	95% CI upper	MERRA-2 RMSE (μg m−3)	MERRA-2 MAE (μg m−3)	n
Winter	h11	0.2743	0.2426	-0.0317	-0.1082	0.0526	42.459	28.917	2142
Winter	h12	0.2557	0.2425	-0.0132	-0.0573	0.0396	42.461	29.791	2142
Winter	h13	0.1750	0.1476	-0.0274	-0.0804	0.0453	45.042	31.202	2142
Winter	h14	0.1647	0.1604	-0.0043	-0.0502	0.0463	44.705	31.918	2142
Winter	h15	0.1355	0.1255	-0.0100	-0.0606	0.0626	45.625	32.515	2142
Winter	h16	0.0910	0.0963	0.0053	-0.0686	0.1007	46.378	32.920	2142
Winter	h17	0.0724	0.0832	0.0108	-0.0463	0.0890	46.714	33.396	2142
Winter	h18	0.0605	0.0684	0.0080	-0.0403	0.0782	47.089	33.474	2142
Winter	h19	0.0532	0.0547	0.0015	-0.0487	0.0735	47.434	34.096	2142
Winter	h20	0.0578	0.0204	-0.0373	-0.0946	0.0289	48.286	34.190	2142
Winter	h21	0.0648	0.0483	-0.0165	-0.0781	0.0599	47.595	33.662	2142
Winter	h22	0.0577	0.0845	0.0268	-0.0325	0.1047	46.680	33.287	2142
Winter	h23	0.0565	0.0705	0.0140	-0.0395	0.0773	47.036	33.639	2142
Winter	h24	0.0769	0.0933	0.0164	-0.0544	0.1150	46.457	33.084	2142

Supplementary Table S6. P90 and P95 warning performance by model and horizon

Thresholds were estimated separately for each horizon from uncapped training targets and applied unchanged to the locked-test period.

Model	Horizon	Rule	Training threshold (µg m−3)	Observed events	Predicted events	Precision	Recall	F1	False alarms	Misses	n
Local-only model	h1	P90	88.8	213	203	0.882	0.840	0.861	24	34	3373
Local-only model	h1	P95	116.4	92	94	0.809	0.826	0.817	18	16	3373
Full model (with MERRA-2)	h1	P90	88.8	213	205	0.883	0.850	0.866	24	32	3373
Full model (with MERRA-2)	h1	P95	116.4	92	87	0.828	0.783	0.804	15	20	3373
Local-only model	h2	P90	88.8	213	191	0.780	0.700	0.738	42	64	3373
Local-only model	h2	P95	116.4	92	70	0.786	0.598	0.679	15	37	3373
Full model (with MERRA-2)	h2	P90	88.8	213	200	0.750	0.704	0.726	50	63	3373
Full model (with MERRA-2)	h2	P95	116.4	92	62	0.758	0.511	0.610	15	45	3373
Local-only model	h3	P90	88.8	213	175	0.691	0.568	0.624	54	92	3373
Local-only model	h3	P95	116.4	92	43	0.651	0.304	0.415	15	64	3373
Full model (with MERRA-2)	h3	P90	88.8	213	169	0.722	0.573	0.639	47	91	3373
Full model (with MERRA-2)	h3	P95	116.4	92	41	0.707	0.315	0.436	12	63	3373
Local-only model	h4	P90	88.8	213	152	0.632	0.451	0.526	56	117	3373
Local-only model	h4	P95	116.4	92	27	0.630	0.185	0.286	10	75	3373
Full model (with MERRA-2)	h4	P90	88.8	213	151	0.682	0.484	0.566	48	110	3373
Full model (with MERRA-2)	h4	P95	116.4	92	38	0.605	0.250	0.354	15	69	3373
Local-only model	h5	P90	88.8	213	142	0.592	0.394	0.473	58	129	3373
Local-only model	h5	P95	116.4	92	35	0.600	0.228	0.331	14	71	3373
Full model (with MERRA-2)	h5	P90	88.8	213	131	0.656	0.404	0.500	45	127	3373
Full model (with MERRA-2)	h5	P95	116.4	92	34	0.441	0.163	0.238	19	77	3373
Local-only model	h6	P90	88.8	213	100	0.580	0.272	0.371	42	155	3373
Local-only model	h6	P95	116.4	92	19	0.526	0.109	0.180	9	82	3373
Full model (with MERRA-2)	h6	P90	88.8	213	80	0.588	0.221	0.321	33	166	3373
Full model (with MERRA-2)	h6	P95	116.4	92	13	0.385	0.054	0.095	8	87	3373
Local-only model	h7	P90	88.8	213	64	0.516	0.155	0.238	31	180	3373
Local-only model	h7	P95	116.4	92	5	0.200	0.011	0.021	4	91	3373
Full model (with MERRA-2)	h7	P90	88.8	213	108	0.519	0.263	0.349	52	157	3373
Full model (with MERRA-2)	h7	P95	116.4	92	25	0.240	0.065	0.103	19	86	3373
Local-only model	h8	P90	88.8	213	70	0.514	0.169	0.254	34	177	3373
Local-only model	h8	P95	116.4	92	5	0.000	0.000	0.000	5	92	3373
Full model (with MERRA-2)	h8	P90	88.8	213	80	0.550	0.207	0.300	36	169	3373
Full model (with MERRA-2)	h8	P95	116.4	92	23	0.174	0.043	0.070	19	88	3373
Local-only model	h9	P90	88.8	213	33	0.515	0.080	0.138	16	196	3373
Local-only model	h9	P95	116.4	92	6	0.000	0.000	0.000	6	92	3373
Full model (with MERRA-2)	h9	P90	88.8	213	96	0.500	0.225	0.311	48	165	3373
Full model (with MERRA-2)	h9	P95	116.4	92	18	0.222	0.043	0.073	14	88	3373
Local-only model	h10	P90	88.8	213	30	0.567	0.080	0.140	13	196	3373
Local-only model	h10	P95	116.4	92	1	0.000	0.000	0.000	1	92	3373
Full model (with MERRA-2)	h10	P90	88.8	213	86	0.500	0.202	0.288	43	170	3373
Full model (with MERRA-2)	h10	P95	116.4	92	14	0.071	0.011	0.019	13	91	3373

Model	Horizon	Rule	Training threshold (µg m−3)	Observed events	Predicted events	Precision	Recall	F1	False alarms	Misses	n
Local-only model	h11	P90	88.8	213	44	0.341	0.070	0.117	29	198	3373
Local-only model	h11	P95	116.4	90	2	0.000	0.000	0.000	2	90	3373
Full model (with MERRA-2)	h11	P90	88.8	213	87	0.414	0.169	0.240	51	177	3373
Full model (with MERRA-2)	h11	P95	116.4	90	1	0.000	0.000	0.000	1	90	3373
Local-only model	h12	P90	88.8	213	16	0.250	0.019	0.035	12	209	3373
Local-only model	h12	P95	116.4	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h12	P90	88.8	213	48	0.438	0.099	0.161	27	192	3373
Full model (with MERRA-2)	h12	P95	116.4	90	2	0.000	0.000	0.000	2	90	3373
Local-only model	h13	P90	88.8	213	21	0.524	0.052	0.094	10	202	3373
Local-only model	h13	P95	116.5	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h13	P90	88.8	213	84	0.429	0.169	0.242	48	177	3373
Full model (with MERRA-2)	h13	P95	116.5	90	7	0.286	0.022	0.041	5	88	3373
Local-only model	h14	P90	88.8	213	8	0.250	0.009	0.018	6	211	3373
Local-only model	h14	P95	116.5	90	1	0.000	0.000	0.000	1	90	3373
Full model (with MERRA-2)	h14	P90	88.8	213	47	0.489	0.108	0.177	24	190	3373
Full model (with MERRA-2)	h14	P95	116.5	90	7	0.000	0.000	0.000	7	90	3373
Local-only model	h15	P90	88.8	213	6	0.167	0.005	0.009	5	212	3373
Local-only model	h15	P95	116.5	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h15	P90	88.8	213	62	0.355	0.103	0.160	40	191	3373
Full model (with MERRA-2)	h15	P95	116.5	90	4	0.000	0.000	0.000	4	90	3373
Local-only model	h16	P90	88.8	213	1	0.000	0.000	0.000	1	213	3373
Local-only model	h16	P95	116.5	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h16	P90	88.8	213	30	0.633	0.089	0.156	11	194	3373
Full model (with MERRA-2)	h16	P95	116.5	90	2	0.000	0.000	0.000	2	90	3373
Local-only model	h17	P90	88.8	213	3	0.667	0.009	0.019	1	211	3373
Local-only model	h17	P95	116.5	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h17	P90	88.8	213	16	0.625	0.047	0.087	6	203	3373
Full model (with MERRA-2)	h17	P95	116.5	90	0	0.000	0.000	0.000	0	90	3373
Local-only model	h18	P90	88.8	213	1	1.000	0.005	0.009	0	212	3373
Local-only model	h18	P95	116.6	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h18	P90	88.8	213	16	0.812	0.061	0.114	3	200	3373
Full model (with MERRA-2)	h18	P95	116.6	90	3	0.000	0.000	0.000	3	90	3373
Local-only model	h19	P90	88.8	213	3	0.333	0.005	0.009	2	212	3373
Local-only model	h19	P95	116.6	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h19	P90	88.8	213	31	0.613	0.089	0.156	12	194	3373
Full model (with MERRA-2)	h19	P95	116.6	90	0	0.000	0.000	0.000	0	90	3373
Local-only model	h20	P90	88.8	213	9	0.222	0.009	0.018	7	211	3373
Local-only model	h20	P95	116.6	90	2	0.000	0.000	0.000	2	90	3373
Full model (with MERRA-2)	h20	P90	88.8	213	35	0.657	0.108	0.185	12	190	3373
Full model (with MERRA-2)	h20	P95	116.6	90	1	0.000	0.000	0.000	1	90	3373
Local-only model	h21	P90	88.6	213	4	0.250	0.005	0.009	3	212	3373
Local-only model	h21	P95	116.6	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h21	P90	88.6	213	38	0.421	0.075	0.127	22	197	3373
Full model (with MERRA-2)	h21	P95	116.6	90	6	0.000	0.000	0.000	6	90	3373
Local-only model	h22	P90	88.6	213	3	0.667	0.009	0.019	1	211	3373

Model	Horizon	Rule	Training threshold (µg m−3)	Observed events	Predicted events	Precision	Recall	F1	False alarms	Misses	n
Local-only model	h22	P95	116.6	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h22	P90	88.6	213	33	0.515	0.080	0.138	16	196	3373
Full model (with MERRA-2)	h22	P95	116.6	90	7	0.143	0.011	0.021	6	89	3373
Local-only model	h23	P90	88.7	213	10	0.100	0.005	0.009	9	212	3373
Local-only model	h23	P95	116.7	90	6	0.000	0.000	0.000	6	90	3373
Full model (with MERRA-2)	h23	P90	88.7	213	22	0.500	0.052	0.094	11	202	3373
Full model (with MERRA-2)	h23	P95	116.7	90	0	0.000	0.000	0.000	0	90	3373
Local-only model	h24	P90	88.6	214	4	0.250	0.005	0.009	3	213	3373
Local-only model	h24	P95	116.7	90	0	0.000	0.000	0.000	0	90	3373
Full model (with MERRA-2)	h24	P90	88.6	214	37	0.459	0.079	0.135	20	197	3373
Full model (with MERRA-2)	h24	P95	116.7	90	1	0.000	0.000	0.000	1	90	3373

Supplementary Table S7A. Locked-test ablation utility and exact-SHAP family shares

Ablation values are full-model R^2 minus R^2 after removal. The detailed ablation values in this table are retained from the frozen Draft 2 secondary package; they are supporting evidence and did not select the Draft 3 primary model. SHAP columns are percentages of total mean absolute attribution at each horizon.

Horizon	MERRA-2 ablation ΔR^2	Engineered local-condition indicator ablation ΔR^2	Calendar/seasonal (%)	Engineered local-condition indicators (%)	Target calendar (%)	Local meteorology (%)	MERRA-2 (%)	Other (%)	Other pollutants (%)	PM2.5 history (%)
h1	-0.0022	0.0045	10.24	7.88	1.18	11.73	5.35	0.00	10.25	53.37
h2	-0.0037	0.0133	13.16	12.63	3.68	12.38	10.59	0.00	14.82	32.75
h3	0.0025	0.0266	15.10	14.38	4.03	12.29	14.28	0.00	13.55	26.36
h4	0.0173	0.0413	17.13	12.47	3.60	14.53	16.25	0.00	15.82	20.19
h5	0.0032	0.0346	13.87	12.01	6.46	13.76	19.55	0.00	15.71	18.64
h6	0.0217	0.0354	14.47	13.24	5.48	13.93	24.19	0.00	18.42	10.29
h7	0.0232	0.0320	13.84	8.90	7.87	18.26	22.72	0.00	14.52	13.89
h8	0.0305	0.0088	17.63	7.65	3.55	18.76	25.12	0.00	14.73	12.56
h9	0.0405	0.0452	16.94	5.43	4.47	19.74	24.19	0.00	19.19	10.05
h10	0.0584	0.0083	15.78	5.19	4.79	19.44	25.31	0.00	20.36	9.12
h11	0.0620	0.0117	17.60	3.58	5.33	19.92	29.36	0.00	15.82	8.39
h12	0.0561	-0.0041	16.64	3.73	5.99	20.39	26.62	0.41	17.54	8.68
h13	0.0870	0.0183	18.36	3.55	4.60	18.85	28.43	0.00	16.73	9.47
h14	0.0905	0.0130	19.52	3.96	4.07	18.71	31.49	0.10	15.75	6.42
h15	0.0917	-0.0182	20.01	4.31	2.96	18.38	32.39	0.08	16.19	5.68
h16	0.0906	-0.0038	18.12	4.40	4.12	18.79	31.24	0.00	17.78	5.56
h17	0.0691	-0.0168	14.20	4.24	6.74	19.33	29.81	0.00	18.83	6.85
h18	0.0911	-0.0475	14.17	3.41	6.96	21.86	29.12	0.00	17.69	6.79
h19	0.1296	0.0109	13.72	3.13	5.40	22.39	28.67	0.00	19.02	7.67
h20	0.0686	-0.0030	13.37	3.42	6.31	20.93	33.79	0.00	16.44	5.75
h21	0.0903	0.0236	17.68	3.45	5.67	20.64	31.79	0.00	13.41	7.35
h22	-0.0013	-0.0038	13.76	3.76	6.15	21.73	32.37	0.00	14.76	7.47
h23	0.0709	0.0406	15.25	6.53	5.34	20.70	30.47	0.00	15.01	6.71
h24	0.0248	-0.0265	12.29	5.55	6.70	23.05	32.31	0.00	14.89	5.20

Supplementary Table S7B. Seasonal MERRA-2 attribution and ablation utility

Seasonal values use fixed-parameter rolling-origin models. Winter values use the coverage-corrected composite.

Season	Horizon	MERRA-2 SHAP share (%)	MERRA-2 ablation ΔR^2	Engineered local-condition indicator ablation ΔR^2
Autumn	h1	4.20	-0.0020	0.0016
Autumn	h2	7.51	0.0017	0.0185
Autumn	h3	11.46	-0.0034	0.0230
Autumn	h4	14.05	0.0071	0.0400
Autumn	h5	15.55	0.0014	0.0452
Autumn	h6	18.12	0.0066	0.0544
Autumn	h7	18.87	-0.0070	0.0352
Autumn	h8	21.76	0.0042	0.0140
Autumn	h9	22.66	0.0308	0.0199
Autumn	h10	23.39	0.0523	0.0092
Autumn	h11	24.87	0.0518	0.0096
Autumn	h12	26.29	0.0242	0.0142
Autumn	h13	26.40	0.0414	0.0040
Autumn	h14	27.04	0.0744	0.0184
Autumn	h15	27.63	0.0866	-0.0029
Autumn	h16	27.82	0.0930	0.0170
Autumn	h17	28.84	0.1118	-0.0021
Autumn	h18	29.53	0.1098	-0.0165
Autumn	h19	28.12	0.0983	-0.0116
Autumn	h20	28.98	0.0738	0.0019
Autumn	h21	30.54	0.0610	0.0001
Autumn	h22	30.38	0.0799	0.0179
Autumn	h23	29.88	0.0639	0.0038
Autumn	h24	29.50	0.0938	0.0288
Spring	h1	5.56	-0.0008	0.0037
Spring	h2	10.01	-0.0031	0.0156
Spring	h3	13.66	0.0071	0.0187
Spring	h4	17.00	0.0063	0.0181
Spring	h5	20.95	0.0053	0.0284
Spring	h6	23.44	-0.0043	0.0204
Spring	h7	24.57	0.0166	0.0167
Spring	h8	25.27	-0.0083	0.0142
Spring	h9	28.40	-0.0018	0.0177
Spring	h10	30.61	-0.0217	-0.0233
Spring	h11	30.53	-0.0130	-0.0053
Spring	h12	30.43	-0.0126	0.0032
Spring	h13	31.73	-0.0092	-0.0064
Spring	h14	33.50	-0.0314	0.0256
Spring	h15	33.92	0.0086	0.0215
Spring	h16	34.60	-0.0105	-0.0120

Season	Horizon	MERRA-2 SHAP share (%)	MERRA-2 ablation ΔR²	Engineered local-condition indicator ablation ΔR²
Spring	h17	32.38	-0.0066	0.0216
Spring	h18	32.36	-0.0063	0.0173
Spring	h19	33.70	-0.0071	-0.0287
Spring	h20	32.27	0.0352	-0.0037
Spring	h21	34.30	0.0402	0.0078
Spring	h22	34.02	0.0368	-0.0146
Spring	h23	29.77	0.0314	0.0080
Spring	h24	30.32	0.0516	0.0156
Summer	h1	3.84	0.0007	-0.0008
Summer	h2	8.38	-0.0008	-0.0003
Summer	h3	10.99	0.0052	0.0046
Summer	h4	12.84	-0.0029	-0.0004
Summer	h5	14.84	0.0068	-0.0083
Summer	h6	17.65	0.0028	0.0043
Summer	h7	19.59	0.0055	-0.0046
Summer	h8	20.54	0.0003	0.0015
Summer	h9	22.42	0.0167	0.0025
Summer	h10	23.75	0.0157	-0.0030
Summer	h11	21.57	0.0133	-0.0216
Summer	h12	21.98	0.0575	0.0139
Summer	h13	23.59	0.0407	-0.0089
Summer	h14	24.79	0.0418	-0.0042
Summer	h15	24.31	0.0172	0.0142
Summer	h16	25.07	0.0324	-0.0017
Summer	h17	26.54	0.0405	-0.0287
Summer	h18	26.34	0.0500	0.0073
Summer	h19	25.00	0.0232	-0.0029
Summer	h20	26.46	0.0320	0.0185
Summer	h21	29.27	0.0274	-0.0087
Summer	h22	24.68	0.0339	0.0103
Summer	h23	25.71	0.0232	0.0103
Summer	h24	25.31	0.0034	-0.0239
Winter	h1	5.88	-0.0141	-0.0002
Winter	h2	10.41	-0.0051	0.0019
Winter	h3	13.74	-0.0055	0.0083
Winter	h4	16.70	-0.0090	0.0196
Winter	h5	19.11	-0.0090	0.0041
Winter	h6	21.44	0.0069	0.0082
Winter	h7	23.20	0.0130	0.0059
Winter	h8	24.09	0.0113	-0.0007
Winter	h9	26.65	0.0040	0.0165
Winter	h10	26.54	0.0086	-0.0051
Winter	h11	26.92	-0.0317	-0.0369
Winter	h12	27.88	-0.0132	0.0303
Winter	h13	27.71	-0.0274	-0.0155

Season	Horizon	MERRA-2 SHAP share (%)	MERRA-2 ablation ΔR²	Engineered local-condition indicator ablation ΔR²
Winter	h14	28.71	-0.0043	0.0056
Winter	h15	28.51	-0.0100	0.0081
Winter	h16	27.71	0.0053	-0.0275
Winter	h17	29.17	0.0108	0.0213
Winter	h18	27.51	0.0080	-0.0107
Winter	h19	28.30	0.0015	-0.0223
Winter	h20	28.66	-0.0373	-0.0217
Winter	h21	29.00	-0.0165	-0.0145
Winter	h22	28.71	0.0268	0.0263
Winter	h23	28.26	0.0140	-0.0202
Winter	h24	29.08	0.0164	0.0021

Supplementary Table S8. Frozen MERRA-2 spatial and temporal alignment sensitivities

All scenarios reused the primary full-model hyperparameters. Scenario roles were frozen before evaluation, keeping the primary model unchanged.

Scenario	Mean R^2	Mean RMSE (μg m−3)	Mean MAE (μg m−3)	Mean bias (μg m−3)	ΔR^2 vs primary	ΔRMSE vs primary	ΔMAE vs primary
primary availability-aligned area weighted cst hour end	0.4473	22.512	15.195	1.797	0.0000	0.0000	0.0000
spatial sensitivity nearest cell cst hour end	0.4434	22.582	15.219	1.715	-0.0038	0.0700	0.0248
retrospective timing sensitivity area weighted cst hour start	0.4476	22.512	15.183	1.816	0.0004	-0.0003	-0.0116
retrospective combined sensitivity nearest cell cst hour start	0.4496	22.473	15.170	1.941	0.0024	-0.0386	-0.0242

Supplementary Table S9A. Prospectively specified feature-reduction acceptance audit

The prospectively specified replacement required all three criteria to pass at every horizon. Seventeen of 24 horizons passed jointly, so the full 222-predictor model remained primary.

Horizon	Band	Reduced predictors	ΔR²	Relative RMSE change	Relative MAE change	R² pass	RMSE pass	MAE pass	Joint pass
h1	h1–h6	119	-0.0024	0.0112	0.0166	Yes	No	No	No
h2	h1–h6	119	0.0073	-0.0148	-0.0086	Yes	Yes	Yes	Yes
h3	h1–h6	119	0.0049	-0.0068	-0.0106	Yes	Yes	Yes	Yes
h4	h1–h6	119	0.0092	-0.0101	-0.0048	Yes	Yes	Yes	Yes
h5	h1–h6	119	0.0134	-0.0128	0.0036	Yes	Yes	Yes	Yes
h6	h1–h6	119	0.0001	-0.0001	0.0031	Yes	Yes	Yes	Yes
h7	h7–h12	130	-0.0099	0.0082	-0.0022	Yes	Yes	Yes	Yes
h8	h7–h12	130	-0.0120	0.0095	0.0147	No	Yes	No	No
h9	h7–h12	130	0.0127	-0.0096	0.0047	Yes	Yes	Yes	Yes
h10	h7–h12	130	0.0096	-0.0071	-0.0163	Yes	Yes	Yes	Yes
h11	h7–h12	130	0.0008	-0.0006	-0.0139	Yes	Yes	Yes	Yes
h12	h7–h12	130	-0.0092	0.0066	-0.0074	Yes	Yes	Yes	Yes
h13	h13–h24	140	0.0017	-0.0012	0.0074	Yes	Yes	Yes	Yes
h14	h13–h24	140	-0.0067	0.0046	0.0069	Yes	Yes	Yes	Yes
h15	h13–h24	140	-0.0001	0.0001	0.0098	Yes	Yes	Yes	Yes
h16	h13–h24	140	-0.0035	0.0023	0.0114	Yes	Yes	No	No
h17	h13–h24	140	-0.0234	0.0151	0.0207	No	No	No	No
h18	h13–h24	140	-0.0248	0.0159	0.0081	No	No	Yes	No
h19	h13–h24	140	0.0022	-0.0014	-0.0022	Yes	Yes	Yes	Yes
h20	h13–h24	140	-0.0049	0.0031	0.0012	Yes	Yes	Yes	Yes
h21	h13–h24	140	0.0004	-0.0003	-0.0066	Yes	Yes	Yes	Yes
h22	h13–h24	140	0.0076	-0.0048	-0.0093	Yes	Yes	Yes	Yes
h23	h13–h24	140	-0.0191	0.0123	0.0110	No	No	No	No
h24	h13–h24	140	-0.0193	0.0123	0.0094	No	No	Yes	No

Supplementary Table S9B. Post-hoc 59-predictor validation profile

The 59-predictor budget remains validation-only and requires new-period testing. For ΔR^2, negative values mean the reduced model was below the full model.

Horizon	Predictors	Reduced model minus full model ΔR^2	Relative RMSE change	Relative MAE change	Post-hoc 5% gate
h1	59	-0.0041	0.0188	0.0143	Yes
h2	59	-0.0040	0.0081	-0.0029	Yes
h3	59	-0.0088	0.0121	0.0083	Yes
h4	59	-0.0055	0.0061	0.0026	Yes
h5	59	0.0134	-0.0128	-0.0062	Yes
h6	59	-0.0118	0.0102	0.0052	Yes
h7	59	-0.0076	0.0062	-0.0062	Yes
h8	59	0.0069	-0.0054	-0.0064	Yes
h9	59	-0.0037	0.0028	0.0022	Yes
h10	59	-0.0229	0.0171	0.0388	Yes
h11	59	-0.0086	0.0063	0.0194	Yes
h12	59	-0.0124	0.0088	0.0179	Yes
h13	59	-0.0201	0.0142	0.0345	Yes
h14	59	-0.0120	0.0083	0.0263	Yes
h15	59	0.0033	-0.0022	0.0164	Yes
h16	59	-0.0024	0.0015	0.0154	Yes
h17	59	0.0034	-0.0022	0.0156	Yes
h18	59	0.0019	-0.0012	0.0213	Yes
h19	59	0.0032	-0.0020	0.0128	Yes
h20	59	0.0053	-0.0034	-0.0013	Yes
h21	59	-0.0056	0.0035	0.0284	Yes
h22	59	0.0029	-0.0018	0.0019	Yes
h23	59	-0.0004	0.0003	0.0091	Yes
h24	59	-0.0237	0.0151	0.0302	Yes

Supplementary Table S9C. Post-hoc 59-predictor stress-test intervals and pass frequencies

Paired 28 h moving-block bootstrap intervals and 30 deterministic-seed refits were evaluated at h10, h13, and h24.

Horizon	Metric	Reduced model minus full model difference	95% CI lower	95% CI upper	CI excludes zero	Bootstrap joint-pass frequency	Seed joint-pass frequency	Seeds passing (of 30)
h10	MAE	0.3880	0.1767	0.5806	Yes	0.833	0.933	28
h10	R2	-0.0238	-0.0549	-0.0010	Yes	0.833	0.933	28
h10	RMSE	0.2749	0.0125	0.5716	Yes	0.833	0.933	28
h13	MAE	0.3455	0.1064	0.5891	Yes	0.901	1.000	30
h13	R2	-0.0209	-0.0521	0.0005	No	0.901	1.000	30
h13	RMSE	0.2300	-0.0076	0.5072	No	0.901	1.000	30
h24	MAE	0.3392	-0.0090	0.6673	No	0.893	0.967	29
h24	R2	-0.0238	-0.0561	0.0059	No	0.893	0.967	29
h24	RMSE	0.2578	-0.0647	0.5543	No	0.893	0.967	29

Supplementary Table S9D. Feature-reduction robustness-tier counts

Counts summarize the complete 666 predictor-band records (222 predictors across three horizon bands) from the prospectively specified stability analysis.

Band	Feature family	Robustness tier	Predictor-band records
h13–h24	Calendar/seasonal timing	core_stable_both	9
h13–h24	Calendar/seasonal timing	correlated_redundant_candidate	8
h13–h24	Calendar/seasonal timing	low_evidence_nonredundant_candidate	2
h13–h24	Calendar/seasonal timing	supporting_shap_or_stable_permutation	10
h13–h24	Engineered local-condition indicators	core_stable_both	1
h13–h24	Engineered local-condition indicators	correlated_redundant_candidate	2
h13–h24	Engineered local-condition indicators	low_evidence_nonredundant_candidate	18
h13–h24	Engineered local-condition indicators	supporting_shap_or_stable_permutation	2
h13–h24	Known target-calendar timing	core_stable_both	3
h13–h24	Known target-calendar timing	low_evidence_nonredundant_candidate	4
h13–h24	Known target-calendar timing	supporting_shap_or_stable_permutation	2
h13–h24	Local meteorology	core_stable_both	4
h13–h24	Local meteorology	correlated_redundant_candidate	5
h13–h24	Local meteorology	low_evidence_nonredundant_candidate	12
h13–h24	Local meteorology	supporting_shap_or_stable_permutation	15
h13–h24	MERRA-2 aerosol/reanalysis	core_stable_both	17
h13–h24	MERRA-2 aerosol/reanalysis	correlated_redundant_candidate	11
h13–h24	MERRA-2 aerosol/reanalysis	low_evidence_nonredundant_candidate	1
h13–h24	MERRA-2 aerosol/reanalysis	supporting_shap_or_stable_permutation	44
h13–h24	Other	correlated_redundant_candidate	1
h13–h24	Other pollutant history	core_stable_both	13
h13–h24	Other pollutant history	correlated_redundant_candidate	3
h13–h24	Other pollutant history	low_evidence_nonredundant_candidate	11
h13–h24	Other pollutant history	supporting_shap_or_stable_permutation	10
h13–h24	PM2.5 history	core_stable_both	8
h13–h24	PM2.5 history	low_evidence_nonredundant_candidate	4
h13–h24	PM2.5 history	supporting_shap_or_stable_permutation	2
h7–h12	Calendar/seasonal timing	core_stable_both	8
h7–h12	Calendar/seasonal timing	correlated_redundant_candidate	6
h7–h12	Calendar/seasonal timing	low_evidence_nonredundant_candidate	5
h7–h12	Calendar/seasonal timing	supporting_shap_or_stable_permutation	10
h7–h12	Engineered local-condition indicators	core_stable_both	2
h7–h12	Engineered local-condition indicators	correlated_redundant_candidate	1
h7–h12	Engineered local-condition indicators	low_evidence_nonredundant_candidate	16
h7–h12	Engineered local-condition indicators	supporting_shap_or_stable_permutation	4
h7–h12	Known target-calendar timing	core_stable_both	2
h7–h12	Known target-calendar timing	low_evidence_nonredundant_candidate	4
h7–h12	Known target-calendar timing	supporting_shap_or_stable_permutation	3
h7–h12	Local meteorology	core_stable_both	6
h7–h12	Local meteorology	correlated_redundant_candidate	4

Band	Feature family	Robustness tier	Predictor-band records
h7–h12	Local meteorology	low_evidence_nonredundant_candidate	11
h7–h12	Local meteorology	supporting_shap_or_stable_permutation	15
h7–h12	MERRA-2 aerosol/reanalysis	core_stable_both	16
h7–h12	MERRA-2 aerosol/reanalysis	correlated_redundant_candidate	17
h7–h12	MERRA-2 aerosol/reanalysis	low_evidence_nonredundant_candidate	3
h7–h12	MERRA-2 aerosol/reanalysis	supporting_shap_or_stable_permutation	37
h7–h12	Other	correlated_redundant_candidate	1
h7–h12	Other pollutant history	core_stable_both	12
h7–h12	Other pollutant history	correlated_redundant_candidate	3
h7–h12	Other pollutant history	low_evidence_nonredundant_candidate	15
h7–h12	Other pollutant history	supporting_shap_or_stable_permutation	7
h7–h12	PM2.5 history	core_stable_both	5
h7–h12	PM2.5 history	correlated_redundant_candidate	1
h7–h12	PM2.5 history	low_evidence_nonredundant_candidate	5
h7–h12	PM2.5 history	supporting_shap_or_stable_permutation	3
h1–h6	Calendar/seasonal timing	core_stable_both	7
h1–h6	Calendar/seasonal timing	correlated_redundant_candidate	4
h1–h6	Calendar/seasonal timing	low_evidence_nonredundant_candidate	6
h1–h6	Calendar/seasonal timing	supporting_shap_or_stable_permutation	12
h1–h6	Engineered local-condition indicators	core_stable_both	5
h1–h6	Engineered local-condition indicators	correlated_redundant_candidate	1
h1–h6	Engineered local-condition indicators	low_evidence_nonredundant_candidate	14
h1–h6	Engineered local-condition indicators	supporting_shap_or_stable_permutation	3
h1–h6	Known target-calendar timing	core_stable_both	2
h1–h6	Known target-calendar timing	low_evidence_nonredundant_candidate	4
h1–h6	Known target-calendar timing	supporting_shap_or_stable_permutation	3
h1–h6	Local meteorology	core_stable_both	4
h1–h6	Local meteorology	correlated_redundant_candidate	4
h1–h6	Local meteorology	low_evidence_nonredundant_candidate	13
h1–h6	Local meteorology	supporting_shap_or_stable_permutation	15
h1–h6	MERRA-2 aerosol/reanalysis	correlated_redundant_candidate	23
h1–h6	MERRA-2 aerosol/reanalysis	low_evidence_nonredundant_candidate	13
h1–h6	MERRA-2 aerosol/reanalysis	supporting_shap_or_stable_permutation	37
h1–h6	Other	correlated_redundant_candidate	1
h1–h6	Other pollutant history	core_stable_both	5
h1–h6	Other pollutant history	correlated_redundant_candidate	2
h1–h6	Other pollutant history	low_evidence_nonredundant_candidate	16
h1–h6	Other pollutant history	supporting_shap_or_stable_permutation	14
h1–h6	PM2.5 history	core_stable_both	4
h1–h6	PM2.5 history	low_evidence_nonredundant_candidate	2
h1–h6	PM2.5 history	supporting_shap_or_stable_permutation	8

Supplementary Table S10. Detailed design-aware comparison with selected prior studies

This table preserves the complete metric-level and comparability evidence supporting the compact design-aware comparison in Supplementary Table S13. R and R^2 remain distinct, and unreported metrics are not inferred.

Study (citation)	Study system / location	Model(s) used	Forecast horizon(s) evaluated	Train/test split method	R^2	RMSE	MAE	Comparability note
This study	This manuscript's Tongliao open-pit coal-mine monitoring record, with forecast-origin-aligned MERRA-2 included in the full model.	CatBoost: local-only model (149 predictors) and full model with MERRA-2 (222 predictors); these are not the post-hoc 59-predictor candidate.	Twenty-four separate direct forecasts, h1–h24.	Chronological target-time train/validation/locked-test partition; one archived primary prediction was generated per eligible target and horizon; no post-test model switching occurred.	Mean h1–h24: 0.4125 for the local-only model and 0.4563 for the full model; h1: 0.9467 and 0.9488, respectively.	Mean h1–h24: 23.285 and 22.512 μg m−3; h1: 7.427 and 7.424 μg m−3, respectively.	Mean h1–h24: 15.362 and 15.195 μg m−3; h1: 4.287 and 4.290 μg m−3, respectively.	Reference anchor: uncapped held-out targets, horizon-specific direct models, and performance averaged over all 24 locked-test horizons.
Wang M et al. (2023; Tongliao) [3]	Same source dataset as this manuscript: Tongliao open-pit coal mine, Inner Mongolia, 2020–2021 hourly record.	CNN-BiLSTM-Attention; compared with Lasso, SVR, XGBoost, LSTM, BiLSTM, CNN-LSTM, and CNN-BiLSTM.	24, 48, 72, 96, and 120 h multi-output forecasts; the selected comparison is the 24 h output.	8:1:1 proportional partition; chronology was not stated; Table 6 averages 10 randomly selected 24 h test sequences.	0.957 (Table 6 mean; also restated in the paper's conclusion).	3.384 μg m−3 (Table 6 mean; 24 h).	2.694 μg m−3 (Table 6 mean; 24 h).	Same source record, but selected test sequences, a multi-output formulation, and a different evaluation unit prevent a head-to-head architecture comparison.
Luan et al. (2023) [17]	Different mine/dataset: Haerwusu open-pit coal mine,	Random Forest with sequential Markov-chain error correction (RF-MC).	Sequential concentration estimation/correction; no explicit future forecast lead.	Chronological: first 7 days for RF training, following observations for testing, and the final 300 observations as	Not reported; Pearson r improved from -0.13 to 0.45 for PM2.5 after one-step correction.	7.40 to 2.56 μg m−3 after one-step correction.	8.51 to 2.44 μg m−3 after one-step correction.	Mine-specific and temporally ordered, but it estimates/corrects near-contemporaneous dust rather

Study (citation)	Study system / location	Model(s) used	Forecast horizon(s) evaluated	Train/test split method	R^2	RMSE	MAE	Comparability note
	northwest China; 41,381 observations collected every 20 s over 10 days.			the Markov correction set.				than evaluating h1–h24 future forecasts.
Qi et al. (2020) [18]	Different mine/dataset: Haerwusu open-cut coal mine, Inner Mongolia, China; 5-min particulate and environmental measurements.	Random Forest with particle-swarm hyperparameter optimization (RF-PSO), using meteorology, noise, and 5-min-lag PM concentrations.	Current PM2.5, PM10, and TSP estimated from 5-min-lag inputs; not a future h1–h24 forecast.	Iterative random 70:30 train/test splitting with 20 random repetitions; fivefold cross-validation within training for hyperparameter selection; not chronological.	Not reported as R^2; Pearson correlation coefficients: PM2.5=0.91, PM10=0.84, TSP=0.86.	Not reported in the verified primary PDF for the cited Pearson-correlation comparison.	Not reported in the verified primary PDF for the cited Pearson-correlation comparison.	Pearson R is not R^2. The near-contemporaneous task and iterative random splitting prevent direct comparison with the chronological multi-horizon locked test.
Lu et al. (2021) [19]	Different mine/dataset: Ha'erwusu open-pit coal mine, China; short-interval PM2.5 and environmental observations.	Gradient-boosting machine optimized by particle swarm optimization, with benchmark models reported in the paper.	5, 10, 20, and 40 min, plus 1 h and 2 h ahead.	Random 70:30 splitting repeated 20 times, with fivefold cross-validation; not chronological.	Not reported as R^2; test Pearson R declined from 0.944 at 5 min to 0.921 at 2 h.	No horizon-specific RMSE table was located in the primary PDF.	No horizon-specific MAE table was located in the primary PDF.	An explicit mining multi-horizon comparator, but its random splits, minute-scale leads, and Pearson R are non-interchangeable with hourly chronological R^2.
Li et al. (2021) [20]	Different mine/dataset: Anjialing open-pit coal mine, Pingshuo mining area, China; total suspended particulate concentration.	LSTM-Attention; compared with LSTM and conventional alternatives.	Previous 4 h used to predict the next 4 h.	Chronological: first 1,525 observations for training and remaining 654 for testing.	0.974 for LSTM-Attention.	16.298 μg m−3 for TSP.	13.028 μg m−3 for TSP.	A genuine chronological mine forecast, but it predicts TSP rather than PM2.5 and uses a 4 h multi-output task.
Wang Z et al. (2023; Ha'erwusu) [21]	Different mine/dataset: Ha'erwusu open-pit coal mine, China; daily meteorology and mine-production intensity.	Machine-learning ensemble/optimization scenarios integrating meteorological and production-intensity predictors; CS1 was the reported leading scenario.	Daily dust concentration; no explicit future lead beyond the daily task was identified.	70:30 split with ordering not stated; 10-fold cross-validation and 5,000 Monte Carlo runs were also used.	Not reported as R^2; average CS1 test Pearson R=0.8280.	8.4987 μg m−3 for the average CS1 test result.	6.9730 μg m−3 for the average CS1 test result.	Mining-specific evidence, but it reports Pearson R, uses an order-unspecified split, and evaluates a daily estimation task.
Yang et al. (2024) [22]	Different mine/dataset: Ha'erwusu open-pit coal mine, China; 365 daily observations combining meteorology and production intensity.	Random Forest optimized by six metaheuristics; MOP-CS was the reported best configuration.	Daily dust concentration; no explicit multi-day future lead was identified.	70:30 split with ordering not stated; 10-fold cross-validation and 5,000 Monte Carlo runs were also used.	Not reported as R^2; average PM2.5 test Pearson R=0.83128 for MOP-CS.	11.61107 μg m−3 for the average PM2.5 MOP-CS test result.	9.46563 μg m−3 for the average PM2.5 MOP-CS test result.	Pearson R and unspecified split ordering prevent direct ranking against the chronological Tongliao test.

Study (citation)	Study system / location	Model(s) used	Forecast horizon(s) evaluated	Train/test split method	R^2	RMSE	MAE	Comparability note
Xiao et al. (2025) [14]	Different mine/dataset: North Weijiamao open-pit mine, Ordos, China; 1,532 observations collected every 10 min from 7–18 August 2023.	GJO-VMD-SVM-LSTM decomposition ensemble; compared with SVM, LSTM, ARIMA, LSSVM, and single-model VMD combinations.	Steps corresponding to 10, 20, 40 min, 1 h, and 2 h; the 10-min step was used for the final model comparison.	Chronological sequence split: initial 70% for training and remaining 30% for testing; alternative ratios were also examined.	0.9256 for the final model at the selected 10-min step.	6.3151 in the paper's concentration unit; the results table does not repeat the unit.	4.5820 in the paper's concentration unit; the results table does not repeat the unit.	The forward split is relevant, but the record spans only 11 days and the selected lead is much shorter than h1–h24.
Ngoc et al. (2026; online 2025) [23]	Different mine/dataset: Sin Quyen open-pit copper mine, Vietnam; hourly observations from 9 April to 5 June 2024.	Dynamic-weighted GA-SVR-Ridge hybrid; compared with component and benchmark regressors.	Hourly PM2.5 prediction; no separate multi-horizon lead structure was stated.	80:20 train/test split; temporal ordering was not stated.	0.831 on the test set for DW-GA-SVRR.	11.003 μg m−3 on the test set.	7.635 μg m−3 on the test set.	The two-month record and unstated split ordering limit direct comparison. The formal issue year is 2026 although publication was online in 2025.
Dhandapani et al. (2023) [24]	Different national dataset: ground monitoring collocated with MERRA-2 aerosol and meteorological fields across India.	Random Forest, LightGBM, XGBoost, and an XGBoost-based stacking regressor.	Contemporaneous hourly and daily surface-PM2.5 estimation, not a future forecast lead.	Random: performance was assessed using a randomly selected 10% subset.	0.73 for XGBoost and 0.77 for stacking; hourly regional stacking R^2 was 0.80 east, 0.63 north, and 0.46 central.	Not reported in the accessible primary abstract.	Not reported in the accessible primary abstract.	The closest MERRA-2 predictor comparator, but it is random-split spatial estimation rather than chronological direct forecasting.
Sayeed et al. (2022) [25]	Different national dataset: US EPA AirNow ground monitoring collocated with MERRA-2 aerosol and meteorological variables across the United States.	Random Forest selected from ten algorithms; regional and pooled models were evaluated.	Contemporaneous hourly and daily grid-level PM2.5 estimation, not a future forecast lead.	Initial 70:30 algorithm-selection split, random 10-fold cross-validation, and regional 70:30 splits.	Not reported; Random-Forest test Spearman correlation was 0.65 initially and 0.82 across cross-validation folds.	7.15 μg m−3 initially; 4.83–5.45 μg m−3 across cross-validation test folds.	Not reported.	Closely matches the use of MERRA-2 predictors, but evaluates random-split contemporaneous estimation and reports Spearman correlation rather than R^2.
Ma et al. (2020) [36]	Different urban dataset: Shanghai, China, using observations and WRF-derived meteorological predictors.	Modified XGBoost; compared with WRF and Lasso.	24–48 h output, including analysis of the next 24 hourly values.	Random training/parameter subsets for 2015–October 2017, followed by chronological verification from November 2017 through December 2018.	Not reported for XGBoost as R^2; full-range Pearson R=0.77 and high-PM2.5-subset Pearson R=0.60.	26.1 μg m−3 for the high-PM2.5 subset; no single full-range XGBoost RMSE was located.	Not reported for the cited XGBoost verification result.	The final verification is forward in time, but development used random subsets and the association metric is Pearson R rather than R^2.
Hossen et al. (2025) [37]	Different regional dataset: 18 EPA monitoring stations in the Taipei area, Taiwan.	CNN-LSTM, BiLSTM, LSTM, GRU, ODE-LSTM, PCNN, CCCFC, and TRCFC; the best model changed with	1, 8, 16, 24, 32, 40, 48, 56, 64, and 72 h forecasts.	Chronological: 2014–2019 training and calendar year 2020 testing.	Not reported on the same all-station aggregation; station-specific R^2 values are reported separately.	Best all-station RMSE was 4.31 μg m−3 at h1, 7.34 at h24, and 8.47 at h72.	For those same models and horizons: 3.05, 5.83, and 6.59 μg m−3.	A strong chronological multi-horizon comparator, but the best model changes with horizon and its station-specific R^2 values are not on

Study (citation)	Study system / location	Model(s) used	Forecast horizon(s) evaluated	Train/test split method	R^2	RMSE	MAE	Comparability note
		horizon.						the same aggregation as the error table.
Boser (2024) [31]	Methodological air-quality example demonstrating validation bias.	Environmental machine-learning validation demonstration.	Not applicable; validation-method study rather than a forecast-horizon experiment.	Explicit contrast between improper random splitting and structure-aware spatial/temporal validation.	A poor model's r^2 was 0.09, appeared as 0.73 when Simpson's paradox was ignored, and inflated to 0.82 with improper splitting.	Not reported.	Not reported.	Not a skill benchmark; it directly supports treating random-split and chronological/structured scores as non-interchangeable.
Zaini et al. (2022) [15]	Different urban dataset: Cheras and Batu Muda stations, Kuala Lumpur, Malaysia, 2018–2019 hourly observations.	EEMD-LSTM; compared with EMD-LSTM, EMD-GRU, LSTM, BiLSTM, Seq2Seq LSTM, and CNN-LSTM.	Separate direct forecasts from h1 to h6.	70:30 sequence holdout; 70% training and the remaining 30% forecasting, with no random shuffle described.	Cheras: 0.9780 at h1 to 0.8880 at h6.	Cheras: 4.2083 to 8.8216 µg m−3 from h1 to h6.	Cheras: 2.8190 to 6.0334 µg m−3 from h1 to h6.	A useful direct short-horizon comparator, but it uses a different urban concentration regime, decomposition preprocessing, and only h1–h6.

Table note: For Wang M et al. [3], Table 6 and the paper's conclusion both report the selected 24 h result as R^2=0.957, RMSE=3.384 µg m−3, and MAE=2.694 µg m−3. The abstract instead labels RMSE=8.985 and MAE=6.957 µg m−3 as 24 h results, although identical values occur in the 120 h columns of Table 7; that internally inconsistent abstract summary was therefore excluded. The separate field-verification values were not selected because the verification period and sample size were not fully specified. A 120 h R^2 value reported only through a later secondary citation [16] was also excluded from the quantitative comparison.

Supplementary Table S11. Evaluation-design characteristics of the selected comparison studies

Entries describe what was reported in the verified source and do not constitute a numerical rigor score.

Study	Primary task	Forecast lead	Split design	Evaluation sample	Multi-horizon	Uncertainty/resampling	Comparison class
This study	Direct future PM2.5 forecast	h1–h24	Chronological target-time split	One-time locked test; 3,373 targets per horizon	Yes	Paired 28 h block bootstrap and seasonal rolling-origin evaluation	Direct chronological reference
Wang M et al. (2023; Tongliao) [3]	Multi-output future PM2.5 forecast	24–120 h	8:1:1; chronology not stated	Primary 24 h result: 10 randomly selected test sequences	Yes	Not reported for the selected result	Same data, different protocol
Luan et al. (2023) [17]	Sequential dust estimation and correction	No explicit future lead	Chronological development/test sequence	Final 300 observations used for Markov correction	No	Sequential correction analysis	Chronological near-contemporaneous correction
Qi et al. (2020) [18]	Near-contemporaneous PM estimation	5-min-lag inputs	Random 70:30 repeated 20 times	Random test subsets	No	Fivefold CV within training and repeated random splits	Random estimation
Lu et al. (2021) [19]	Future PM2.5 forecast	5 min-2 h	Random 70:30 repeated 20 times	Random test subsets	Yes	Fivefold CV and repeated random splits	Random short-lead forecast
Li et al. (2021) [20]	Future TSP forecast	Next 4 h	Chronological sequence holdout	First 1,525 training; remaining 654 testing	Multi-output	Not reported	Chronological, different target
Wang Z et al. (2023; Ha'erwusu) [21]	Daily dust estimation	No explicit future lead	70:30; order not stated	Test partition plus cross-validation	No	10-fold CV and 5,000 Monte Carlo runs	Order-unspecified mine estimation
Yang et al. (2024) [22]	Daily PM2.5 estimation	No explicit future lead	70:30; order not stated	365 daily observations	No	10-fold CV and 5,000 Monte Carlo runs	Order-unspecified mine estimation
Xiao et al. (2025) [14]	Future dust forecast	10 min-2 h; final comparison at 10 min	Chronological 70:30 sequence split	1,532 observations over 11 days	Several leads examined	Alternative split-ratio sensitivity	Chronological very-short-lead forecast
Ngoc et al. (2026; online 2025) [23]	Hourly PM2.5 prediction	Separate lead structure not stated	80:20; order not stated	Approximately two months of observations	No	Not reported	Order-unspecified mine forecast
Dhandapani et al. (2023) [24]	Contemporaneous MERRA-2 PM2.5 estimation	Hourly and daily estimation	Random 10% assessment subset	Ground-monitoring/MERRA-2 collocations	Not forecast horizons	Regional evaluation	Random MERRA-2 estimation
Sayeed et al. (2022) [25]	Contemporaneous MERRA-2 PM2.5 estimation	Hourly and daily estimation	Random 70:30 and random 10-fold CV	US AirNow/MERRA-2 collocations	Not forecast horizons	Random CV and regional splits	Random MERRA-2 estimation
Ma et al. (2020) [36]	Future urban PM2.5 forecast	24–48 h	Random development subsets; chronological verification	November 2017–December 2018 verification	Yes	Separate forward verification period	Mixed development/chronological verification
Hossen et al. (2025) [37]	Future regional PM2.5 forecast	h1–h72	Chronological: 2014–2019 train, 2020 test	Full calendar-year test across 18 stations	Yes	Paired t-tests between model RMSE results; no blocked forecast-uncertainty analysis	Regional chronological multi-horizon forecast
Boser (2024) [31]	Environmental validation-method study	Not applicable	Structured versus improper splitting	Methodological air-quality example	Not applicable	Validation-bias demonstration	Methodological evidence
Zaini et al. (2022) [15]	Direct future urban PM2.5 forecast	h1–h6	70:30 sequence holdout	Remaining 30% used for forecasting	Yes	Not reported	Sequence-holdout short-horizon forecast

Table note: 'Not reported' means that the verified source did not provide the corresponding procedure for the selected result. Categories describe task and validation structure; they do not rank model quality.

Supplementary Table S12. Validation-only sensitivity to consistent pressure cleaning

The archived implementation retained three 0.0 kPa observations in direct pressure-derived predictors but masked them for physically constrained engineered indicators. The corrected-pressure sensitivity converted values outside 80–110 kPa to missing before training-only median imputation and all feature construction. Both conditions used the archived horizon-specific CatBoost parameters, original training targets, and original validation targets; the locked test remained untouched.

Quantity	Archived pressure handling	Corrected pressure handling	Corrected minus archived
Local-only mean validation R^2, h1–h24	0.3616	0.3518	-0.0098
Local-only mean validation RMSE (µg m−3), h1–h24	15.509	15.625	+0.116
Local-only mean validation MAE (µg m−3), h1–h24	9.999	10.065	+0.066
Full-model mean validation R^2, h1–h24	0.3694	0.3561	-0.0133
Full-model mean validation RMSE (µg m−3), h1–h24	15.411	15.575	+0.164
Full-model mean validation MAE (µg m−3), h1–h24	9.848	9.939	+0.091
Mean full-minus-local validation ΔR^2, h1–h6	+0.0053	+0.0006	-0.0047
Mean full-minus-local validation ΔR^2, h7–h12	+0.0135	+0.0070	-0.0066
Mean full-minus-local validation ΔR^2, h13–h24	+0.0061	+0.0048	-0.0013
Mean full-minus-local validation ΔR^2, h1–h24	+0.0078	+0.0043	-0.0035
Horizons with positive full-minus-local validation ΔR^2	19/24	12/24	-7 horizons
Maximum absolute horizon-specific ΔR^2 change, local-only	-	-	0.0383
Maximum absolute horizon-specific ΔR^2 change, full model	-	-	0.0354

Table note: The correction changed 104 values across 12 pressure-derived feature columns and 75 origin rows. All 96 planned model fits completed. The latest raw, target, and prediction timestamp was 7 August 2021 18:00, one hour before the locked-test start; protected locked-test prediction and metric hashes were unchanged. Pressure-anomaly handling mattered in validation; because locked-test artifacts were protected, archived locked-test estimates remained unchanged.

Supplementary Table S13. Compact design-aware comparison with selected prior studies

Performance values are reported only when verified from the original publications. Pearson or Spearman correlation coefficients are not equivalent to R^2. The detailed metric-level and evaluation-design evidence appears in Supplementary Tables S10 and S11.

Study	System/task	Model and horizon	Evaluation design	Verified performance values	Comparison class
This study	Tongliao mine; future hourly PM2.5	CatBoost, local-only and full models; 24 direct models, h1–h24	Chronological target-time split; one archived primary prediction per eligible target and horizon; no post-test model switching	Mean R^2 0.4125→0.4563; RMSE 23.142→22.323; MAE 15.331→15.061 μg m−3. h1 R^2 0.9467→0.9488	Reference anchor: genuine multi-horizon forecast
Wang M et al. (2023) [3]	Same Tongliao record; multi-output PM2.5	CNN-BiLSTM-Attention; 24–120 h; selected 24 h output	8:1:1, chronology unstated; Table 6 mean of 10 randomly selected test sequences	R^2 0.957; RMSE 3.384; MAE 2.694 μg m−3	Same data, different protocol/evaluation unit
Luan et al. (2023) [17]	Haerwusu mine; sequential PM2.5 estimation/correction	RF with Markov correction; no explicit future lead	First 7 d training, later observations testing; final 300 for correction	Pearson r −0.13→0.45; reported RMSE 7.40→2.56 and MAE 8.51→2.44 μg m−3	Mine-specific, temporally ordered, near-contemporaneous
Qi et al. (2020) [18]	Haerwusu mine; 5-min particulate estimation	RF-PSO using 5-min-lag inputs; not h1–h24 forecasting	Random 70:30 repeated 20 times; fivefold CV in training	Pearson R: PM2.5 0.91, PM10 0.84, TSP 0.86	Mine-specific, random-split estimation
Lu et al. (2021) [19]	Ha'erwusu mine; short-lead PM2.5	PSO-optimized gradient boosting; 5 min–2 h	Random 70:30 repeated 20 times; fivefold CV	Test Pearson R 0.944 at 5 min to 0.921 at 2 h	Mine-specific, random-split short-lead forecast
Li et al. (2021) [20]	Anjialing mine; future TSP	LSTM-Attention; previous 4 h to next 4 h	Chronological 1,525/654 sequence split	R^2 0.974; RMSE 16.298; MAE 13.028 μg m−3	Mine-specific chronological forecast, different pollutant
Wang Z et al. (2023) [21]	Ha'erwusu mine; daily dust estimation	Optimized ML ensemble; daily task	70:30 ordering unstated; 10-fold CV and 5,000 Monte Carlo runs	Pearson R 0.8280; RMSE 8.4987; MAE 6.9730 μg m−3	Mine-specific, order-unspecified estimation
Yang et al. (2024) [22]	Ha'erwusu mine; daily PM2.5 estimation	Metaheuristic-optimized RF; daily task	70:30 ordering unstated; 10-fold CV and 5,000 Monte Carlo runs	Pearson R 0.83128; RMSE 11.61107; MAE 9.46563 μg m−3	Mine-specific, order-unspecified estimation
Xiao et al. (2025) [14]	North Weijiamao mine; short-lead PM2.5	GJO-VMD-SVM-LSTM; 10 min–2 h; selected 10 min	Chronological 70:30 sequence split; 11-d record	R^2 0.9256; RMSE 6.3151; MAE 4.5820 (paper's concentration unit)	Mine-specific chronological short-lead forecast
Ngoc et al. (2026) [23]	Sin Quyen mine; hourly PM2.5	Dynamic-weighted GA-SVR-Ridge; lead structure unstated	80:20; temporal ordering unstated; two-month record	R^2 0.831; RMSE 11.003; MAE 7.635 μg m−3	Mine-specific, order-unspecified hourly prediction
Dhandapani et al. (2023) [24]	India; MERRA-2 surface-PM2.5 estimation	XGBoost and stacking; contemporaneous hourly/daily	Randomly selected 10% evaluation subset	R^2 0.73 XGBoost, 0.77 stacking; regional 0.80/0.63/0.46	MERRA-2 predictor evidence, random estimation
Sayeed et al. (2022) [25]	United States; MERRA-2 grid PM2.5 estimation	Random Forest; contemporaneous hourly/daily	Random 70:30 and 10-fold CV	Spearman R 0.65 initially, 0.82 across CV; RMSE 7.15 then 4.83–5.45 μg m−3	MERRA-2 predictor evidence, random estimation
Ma et al. (2020) [36]	Shanghai; 24–48 h PM2.5	Modified XGBoost with WRF predictors	Random development subsets; chronological Nov 2017–Dec 2018 verification	Pearson R 0.77 overall and 0.60 at high PM2.5; high-PM2.5 RMSE 26.1 μg m−3	Forward verification after partly random development
Hossen et al. (2025) [37]	Taipei region; future PM2.5 across 18 stations	ODE-based and recurrent neural models; h1–h72	Chronological 2014–2019 training and calendar-year 2020 testing	Best all-station RMSE/MAE: 4.31/3.05 at h1, 7.34/5.83 at h24, and 8.47/6.59 μg m−3 at h72; best model changed by horizon	Regional chronological multi-horizon forecast
Boser (2024) [31]	Air-quality validation-method example	Validation-bias demonstration; no forecast horizon	Improper random splitting contrasted with structure-aware validation	True r^2 0.09 appeared as 0.73 and inflated to 0.82	Methodological evidence; not a skill benchmark
Zaini et al. (2022) [15]	Kuala Lumpur; future hourly PM2.5	EEMD-LSTM; direct h1–h6	70:30 sequence holdout; no shuffle described	R^2 0.9780→0.8880; RMSE 4.2083→8.8216; MAE 2.8190→6.0334 μg m−3	Urban chronological short-horizon forecast

Table note: For Wang M et al. [3], Table 6 and the paper's conclusion both report the selected 24 h values shown above. The abstract instead labels RMSE=8.985 and MAE=6.957 μg m−3 as 24 h results, although identical values occupy the 120 h columns of Table 7; those internally inconsistent abstract values were excluded. The separate field-verification summary and a 120 h R^2 reported only through a later secondary citation [16] were also excluded from the primary comparison.